Price Discrimination, Consumer Search, and Online Sales in the Automobile Industry

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May 14, 2026

Abstract. We investigate the welfare consequences of introducing an online distribution channel in the French car industry, where most sales take place in person at discounts off list prices. We estimate a model with unobserved third-degree price discrimination and sequential consumer search associated with visiting car dealers. For our counterfactual analysis, we develop a model of multi-channel sequential search and introduce an online distribution channel that charges list prices but entails lower search costs. The online channel increases competition and leads to lower list prices. For consumers more sensitive to prices than to search costs, firms still offer in-person discounts. For those more sensitive to search costs, firms raise in-person prices to steer them toward the online channel. The online channel leads to market expansion and is profitable for firms, with profitability rising with search cost reductions. However, the costs and benefits of the online channel are unevenly distributed among consumers, with older and wealthier consumers obtaining most of the benefits.

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We would like to thank for helpful comments and suggestions Hendrik Döpper, Pierre Dubois, Gaston Illanes, Andre Veiga, and the participants of various seminars and conferences. We acknowledge financial support from the European Research Council under grant ERC-2019-STG-852815 "PRIDISP" and the Agence Nationale de Recherche under grant ANR-17-EUR-0010 (Investissements d'Avenir program). Access to some confidential data, on which is based this work, has been made possible within a secure environment offered by CASD – Centre d'accès sécurisé aux données (Ref. 10.34724/CASD).

1 Introduction

Technological progress has facilitated online transactions for many products and services. Doing business online has several advantages for firms and consumers. On the one hand, firms may gain access to a larger consumer base and may save on the costs of establishing and maintaining a dense network of physical stores. Consumers, on the other hand, may benefit from having access to a wider variety of products and services and avoiding part of the search costs associated with visiting physical stores for their purchases.

As consumers and firms are getting used to online marketplaces, there is evidence suggesting that even large and expensive products, such as cars, will be traded online. A pioneering example is Tesla, which operates almost exclusively online. Ordering a vehicle, signing the contract, and making the payment all happen through the company's website, and home delivery at no extra cost is available within 354km (220 miles) of a Tesla distribution center. Similarly, Ford's CEO Jim Farley announced in 2022 a plan to move part of the downstream activity online, ending the traditional dealership model and selling directly to consumers at list prices.[1] Moving distribution online at list prices and simplifying the transaction process are part of a larger plan to enforce price transparency and improve consumer convenience.[2] Other manufacturers are expected to follow suit if the examples of Tesla and Ford prove successful.

It is well documented that consumers often obtain discounts off list prices when buying a new car in person at a car dealer (D'Haultfœuille et al., 2019; Goldberg, 1996; Langer, 2011; Sagl, 2024). Through personal interactions, car dealers observe consumers' characteristics and then may offer discounts off list prices (or valuable advantages like free upgrades or an extended warranty). We interpret these discounts as a form of third-degree price discrimination. In addition, purchasing a car in person involves costly search for consumers, due to the need to travel to car dealers and the associated opportunity cost of time (Moraga-González et al., 2023; Nurski and Verboven, 2016).

A hypothetical online distribution channel would introduce a trade-off for consumers. On the one hand, by choosing the online distribution channel, consumers would forgo potential discounts and face list prices. On the other hand, conducting part of the transaction online would allow consumers to save on search costs by avoiding some or even all visits to car dealers. Because of this trade-off, car manufacturers' incentives to price discriminate in the in-person channel may be altered by the presence of an online

[1] Source: Phoebe Ward Howard, "Ford CEO Farley says electric vehicles will be sold 100% online, have non-negotiable price", Detroit Free Press. The full article is available here.

[2] Additional information can be found on Ford's website, see here.

distribution channel. For example, they may stop offering in-person discounts and divert sales online if consumers are mainly interested in saving on search costs. In this paper, we investigate this trade-off and evaluate the welfare consequences of introducing an online distribution channel in the French car industry.

Using municipal-level data on the French car industry during 2009–2021—a period in which most transactions took place in person at car dealers—we estimate a model of in-person car sales with unobserved third-degree price discrimination and sequential consumer search. Our model combines insights from D'Haultfœuille et al. (2019) and Moraga-González et al. (2023). As in D'Haultfœuille et al. (2019), recognizing that consumers often receive discounts while only list prices are observed, our model treats unobserved discounts as additional parameters to be estimated using equilibrium pricing conditions. As in Moraga-González et al. (2023), recognizing that different municipalities are more or less conveniently located in the network of car dealers, our model also allows for sequential consumer search across car dealers.

To estimate the model, we combine data on car sales—by age and municipality—car characteristics and list prices, municipal-level demographics, car dealer locations, and consumer driving distances. We categorize consumers into demographic groups based on their age and the median income in their municipality. We assume that car dealers observe these groups and engage in third-degree price discrimination. We augment aggregate demand- and supply-side moments with micro moments that are informative about unobserved preferences and search costs. For example, we match group-specific observed average distances with the corresponding model predictions. These micro moments not only carry information about search costs but are also robust to the potential endogeneity of distance without requiring additional instruments.[3]

Given our model estimates and a novel survey of attitudes toward online car purchases, we simulate the introduction of an online distribution channel and study its interactions with the traditional in-person distribution channel. We extend our model to enable multi-channel sequential search: consumers first choose whether to search each car model in person or online, and then search sequentially given the chosen search channels (with different car models potentially searched through different channels). In counterfactuals, we assume that firms charge list prices online while offering discounts for in-person transactions.[4] We also assume that consumers face reduced search costs when

[3]Distance could be endogenous if, for example, car dealers took into account unobserved preferences at the time of choosing their location.

[4]Following the stated intentions of firms and the practice of Tesla (see discussion above), we favor this assumption over the alternative that firms price discriminate also in the online channel. However, for

searching online. Since car dealers are important for test drives, after-sale services, and maintenance, we believe that search costs may still matter when shopping online, but to a lesser extent than for in-person transactions. We then consider various levels of search cost reductions for the online channel, using as a baseline the 37% reduction that we estimate from the launch of the Tesla Model 3 in 2019.

Our counterfactuals reveal that, through a reduction in search costs, the online channel increases competition in the industry and leads to lower list prices (what we call the *competitive effect*). Firms' pricing responses are heterogeneous across consumer groups. For consumers more sensitive to prices than to search costs, firms still offer in-person discounts. For consumers more sensitive to search costs instead, firms raise in-person prices to divert them toward the online channel, where the absence of discounts makes each sale more profitable (what we call the *online steering effect*). The online channel thus becomes especially attractive for consumers who were already receiving small or no in-person discounts, typically older and wealthier consumers who represent the majority of car buyers in France. As a result, while some younger and less wealthy consumers purchase fewer cars, we find a transfer of sales from the in-person to the online channel and an overall market expansion.

We find that price discrimination, taken in isolation, benefits only some consumers, typically the younger and the less wealthy, as they receive the largest discounts off list prices. In aggregate, the effect of price discrimination on consumer surplus is a small loss, while its effect on industry profit is a gain of 3.3%. In contrast, search costs are detrimental to all consumers and firms. Reducing search costs by 37% (the estimated reduction for Tesla's online channel) already generates a larger increase in consumer surplus than that implied by eliminating price discrimination. It also generates an increase in industry profit that is around 70% of that implied by price discrimination. Eliminating search costs altogether yields welfare benefits around four times as large as those of price discrimination for consumers and twice as large for firms, meaning that price discrimination plays a somewhat less prominent role than search costs in this industry.

The online distribution channel is generally profitable for car manufacturers despite the average decrease in list prices, as it leads to market expansion. Industry profit increases by up to 3.3%, and the increase generally grows with the reduction in search costs. In terms of consumer surplus, however, the online channel gives rise to heterogeneous

completeness, we perform a set of counterfactuals in which firms price discriminate in both distribution channels, see Appendix Appendix D.

effects with winners and losers. Consumers who tend to benefit from the introduction of an online channel are older and wealthier. These consumers would typically not receive in-person discounts in any case, and thus the search cost savings (in addition to the lower list prices) make them unambiguously better off. For other consumers, the introduction of an online channel can bring either losses or gains, depending on the scenario.

We assume that the network of car dealers remains fixed when the online distribution channel is introduced. Although potential adjustments may be important, we maintain this assumption because of the practical infeasibility of also incorporating a layer of endogenous network formation into an already rich structural model of unobserved third-degree price discrimination and multi-channel sequential consumer search. In this sense, our results should be interpreted as a collection of short-run responses to the introduction of an online distribution channel. However, to provide some insight into this mechanism, we also consider counterfactuals in which the 5%, 10%, and 20% lowest performing car dealers of each brand exit the market. Our results suggest that, when the network of car dealers becomes less dense—a mechanism akin to an increase in search costs—the online channel must generate larger search cost savings in order to induce market expansion.

In a final set of counterfactuals, we allow for the possibility that the online channel enables car manufacturers to save on marginal costs, which could happen when bypassing the "middleman" (e.g., removal of double marginalization and extra inventory costs). When firms' "effective" marginal costs decrease, as expected, equilibrium prices also decrease, overall car sales increase, and as a consequence, both consumer surplus and industry profit increase. Importantly, the fact that in this case industry profit increases means that there could be ways of redistributing it so as to keep car dealers at least as well off as in the scenario without an online distribution channel.

Related literature. Our paper proposes a novel structural model of unobserved third-degree price discrimination and multi-channel sequential consumer search. Our model is closely related to recent empirical approaches proposed to study price discrimination (e.g., D'Haultfœuille et al., 2019; Iaria and Wang, 2024; Sagl, 2024). For example, Sagl (2024) studies the trucking industry in Texas and finds that most of the observed price dispersion can be explained by consumer unobservables (or soft information), as opposed to demographics. His analysis leverages consumer-level transaction prices and repeated purchases over time. In the different context of France, D'Haultfœuille et al. (2019) instead rely on list prices and recover unobserved transaction prices resulting from third-degree price discrimination based on consumer demographics. Our model extends D'Haultfœuille et al. (2019) to provide a unified framework that incorporates (potentially unobserved) transaction prices and sequential consumer search and that can be used to investigate the relationship between the two in oligopolistic industries with differentiated products.

Our model also contributes to the empirical literature on consumer search (e.g., Moraga-González et al., 2023; Murry and Zhou, 2020; Yavorsky et al., 2021). In the context of the car industry, this literature deals with situations in which consumers need to personally visit car dealers to learn about some of the features of car models and where search costs depend on the distance to car dealers. Our model extends the sequential consumer search model by Moraga-González et al. (2023) in two ways. First, it incorporates into the framework unobserved third-degree price discrimination as in D'Haultfœuille et al. (2019). Second, our model extends Moraga-González et al. (2023) from single- to multi-channel sequential consumer search: in our counterfactuals, consumers first choose which channel to use for their search (separately for each product) and then perform their sequential search as in Moraga-González et al. (2023). Importantly, although in our context there are only two potential search channels consumers can choose from (in-person and online), the model can easily be extended to any number of search channels.

Beyond these methodological aspects, our paper contributes to several other branches of the literature. First, it relates to a growing literature on the welfare effects of e-commerce, such as Brown and Goolsbee (2002) on the impact of comparison websites on insurance prices in the US and Morton et al. (2001) on car referral websites (a precursor of online sales in the car industry). Similarly to Pozzi (2013), Fan et al. (2018), and Forman et al. (2009), our findings show that the coexistence of an in-person and an online distribution channel can generate welfare gains through both increased competition and reductions in search costs. However, they also highlight important distributional effects, showing that firms and the most inelastic consumers can obtain most of the benefits. Along the lines of Huang and Bronnenberg (2023) and Brynjolfsson et al. (2003), our analysis also illustrates that these forces are closely related to the variety of products effectively available to consumers, revealing novel mechanisms through which an online channel can limit the ability of brick-and-mortar stores to price discriminate.

Second, we contribute to the literature on price discrimination and price dispersion in retail markets. Seminal work by Corts (1998) and Thisse and Vives (1988) and recent work by Iaria and Wang (2024) and Rhodes and Zhou (2024) suggest a complex relationship between price discrimination and welfare, particularly in oligopolistic industries.

Previous empirical studies on the car industry investigated price discrimination based on consumer demographics; see Ayres and Siegelman (1995), Goldberg (1996), Harless and Hoffer (2002), and Chandra et al. (2017). These studies find contrasting evidence linking price dispersion to demographics (typically gender and race). We provide novel evidence on the relationship among transaction prices, income, and age (we find no relationship with gender), and more broadly on the relationship between price discrimination and spatial differentiation in an oligopolistic industry with differentiated products (for related evidence with homogeneous products, see Miller and Osborne, 2014).

Third, our work relates to recent papers studying price personalization in online markets. Shiller (2020) studies price personalization for Netflix subscription plans based on browsing histories, while Dubé and Misra (2023) study price personalization for a digital firm based on observable consumer characteristics. These studies leverage the vast amount of information available online to investigate the consequences of price personalization for a firm. Instead, we focus on the consequences of online sales for an oligopolistic industry with differentiated products and a long tradition of brick-and-mortar stores. The stated intention of car manufacturers is to use the online channel to enforce price transparency rather than personalization, allowing us to explore a different type of interaction between the online and in-person channels.

Finally, our work is closely related to Duch-Brown et al. (2023), who study the interaction between online and in-person sales in the portable PC industry in Europe. In their application, price dispersion occurs in the online market as a result of geoblocking restrictions on cross-border transactions. They show that banning these restrictions results in the convergence of prices to a unique European-level price for each product sold online. Our findings reveal a similar convergence mechanism. However, our results also show that when there is consumer heterogeneity in price and search cost sensitivities, firms also continue to price discriminate in the in-person channel. More broadly, we contribute to Duch-Brown et al. (2023) by specifically investigating the roles of spatial differentiation and search costs in the transition to market integration promoted by an online distribution channel.

2 Baseline Model

We combine the models of sequential consumer search by Moraga-González et al. (2023) and of unobserved (to the researcher) third-degree price discrimination by D'Haultfœuille et al. (2019). Consumers do not fully know the indirect utility they would obtain from a car model until they visit a car dealer that sells it. In addition, each consumer belongs to one of D mutually exclusive groups based on their observed demographics and, for any car model j, producers price discriminate by offering a potentially different transaction price p_{jd} to the consumers of each group d. In our sample period in France, the only known exception to third-degree price discrimination was Tesla: for any car model j sold by Tesla, we assume throughout that $p_{jd} = \overline{p}_j$.

2.1 Consumer search

Consider consumer i belonging to demographic group d (an age and income group) and living in municipality m (a town in France). We omit the time subscript for simplicity. Their indirect utility from purchasing car model $j \in \mathcal{J} \equiv \{1, ..., J\}$ is

$$U_{ijdm} = \underbrace{X_j'\beta_d + \alpha_d p_{jd} + \xi_{jd}}_{\delta_{jd}} + \underbrace{X_j'(\Pi_d \mathrm{dem}_{dm} + \Sigma_d \nu_i)}_{\mu_{jdm}(\nu_i)} + \epsilon_{ijdm}, \tag{1}$$

where X_j is a vector of observed car characteristics that is invariant across groups (e.g., horsepower), p_{jd} is the transaction price faced by group d, and ξ_{jd} is the residual portion of average indirect utility of group d. Note that neither p_{jd} nor ξ_{jd} varies across municipalities.[5] The term $\mu_{jdm}(\nu_i)$ captures individual-level deviations in the preferences for X_j from the group average $X_j'\beta_d$. The vector dem_{dm} collects observable average demographics (e.g., income, household size, an urban indicator) specific to the individuals of group d living in municipality m, while ν_i is a vector of random coefficients.[6] Variable ϵ_{ijdm} is an idiosyncratic error term assumed to be i.i.d. Gumbel$(0, 1)$. The indirect utility of the outside option (not purchasing any car model in $\mathcal{J}$) is $U_{i0dm} = \epsilon_{i0dm}$. As in D'Haultfœuille et al. (2019), the parameters $(\beta_d, \alpha_d, \Pi_d, \Sigma_d)$ are allowed to be heterogeneous across demographic groups.

Following Moraga-González et al. (2023), consumers do not know ex-ante the realization of their ϵ_{ijdm}'s and must visit car dealers to find out. We define a search as a trip from home to a single car dealer, and we assume that consumers learn the ϵ_{ijdm}'s of all car models available at the visited car dealer. In particular, each producer $f \in \mathcal{F}$ sells a disjoint set of products $\mathcal{G}_f \subset \mathcal{J}$ with $\mathcal{G}_f \cap \mathcal{G}_g = \emptyset$ for $f \neq g$. Every producer f has a network of car dealers and each of f's car dealers is assumed to sell all of f's car models $\mathcal{G}_f$. When searching for the car models of producer f, consumers are assumed to visit the closest car dealer to their municipality selling $\mathcal{G}_f$. To visit the closest car dealer of

[5] As we discuss in Section 3, this restriction relates to data availability and identification.

[6] We also investigated specifications with heterogeneous coefficients on price within each demographic group d, but we did not obtain qualitatively different results. Given the higher computational complexity of these specifications (more on this below), we opted for the simpler indirect utility (1).

producer f, consumer i of group d living in municipality m must incur a search cost sc_{ifdm}. After the visit, the consumer learns ϵ_{ijdm} for all $j \in \mathcal{G}_f$.

Search is sequential and ordered, and consumers have costless recall. After each visit to a car dealer, consumers decide whether to buy any of the inspected car models so far, to opt for the outside option, or to continue searching. Before searching, consumer i of group d and living in m knows: (a) the location of the closest car dealer of each producer and the subset of car models each sells, (b) all elements of indirect utility (1)—including p_{jd} and ξ_{jd}—but not ϵ_{ijdm} for each j, (c) ϵ_{i0dm} and the distribution of all ϵ_{ijdm}'s, (d) the search cost sc_{ifdm} of visiting the closest car dealer of producer f, for all $f \in \mathcal{F}$.

Given these assumptions, Moraga-González et al. (2023) show that the Weitzman rule characterizes optimal consumer search (Weitzman, 1979): consumers visit car dealers in descending order of reservation value and stop as soon as the best option found so far exceeds the next reservation value. The reservation value r_{ifdm} of consumer i of group d in municipality m for the closest of f's car dealers is the unique r satisfying $H_0(r - \delta_{ifdm}) = \mathrm{sc}_{ifdm}$, where

$$
\begin{aligned}
H_0(r) &\equiv \int_r^\infty (z - r) dG(z), \; G \text{ is a Gumbel}(0,1) \text{ c.d.f., and} \\
\delta_{ifdm} &\equiv \ln \left(\sum_{h \in \mathcal{G}_f} \exp(\delta_{hd} + \mu_{hdm}(\nu_i)) \right), \text{ so that} \\
r_{ifdm} &= \delta_{ifdm} + H_0^{-1}(\mathrm{sc}_{ifdm})
\end{aligned}
\tag{2}
$$

with $H_0 : \mathbb{R} \to (0, \infty)$ a strictly decreasing function representing $\mathbb{E}[\max\{\epsilon - r, 0\}] = \mathbb{E}[(\epsilon - r) \cdot \mathbb{1}\{\epsilon > r\}]$ for ϵ distributed Gumbel$(0,1)$. Following Moraga-González et al. (2023) and Armstrong (2017); Choi et al. (2018), the purchase decision resulting from the Weitzman rule is equivalent to a two-step protocol that does *not* involve any search. Define

$$
\mathrm{w}_{ifdm} \equiv \min \left\{ r_{ifdm}, \max_{j \in \mathcal{G}_f} \{U_{ijdm}\} \right\}. \tag{3}
$$

In the first step, consumer (i, d, m) observes w_{ifdm} for all $f \in \{0\} \cup \mathcal{F}$ (where 0 denotes the outside option) and picks the f with the highest w_{ifdm}. In the second step, given f, she observes U_{ijdm} for all $j \in \mathcal{G}_f$ and picks the j with the highest U_{ijdm}.

2.2 Purchase probabilities and market shares

According to the two-step procedure described above, the probability with which consumer (i, d, m) purchases car model j of producer f can be expressed as:

$$\begin{aligned}\Pr_{idm}(j) &= \Pr_{idm}(j \mid f) \cdot \Pr_{idm}(f) \\ &= \Pr_{idm}\left(U_{ijdm} \geq \max_{h \in \mathcal{G}_f}\{U_{ihdm}\}\right) \cdot \Pr_{idm}\left(\mathrm{w}_{ifdm} \geq \max_{g \in \{0\} \cup \mathcal{F}}\{\mathrm{w}_{igdm}\}\right).\end{aligned} \tag{4}$$

Consumers observe all w_{igdm}'s and U_{ihdm}'s and thus make deterministic choices; the probabilities in (4) arise from the researcher's perspective, who does not observe the realizations of ϵ_{ijdm} and w_{ifdm}. To obtain a closed-form expression for (4), we follow Moraga-González et al. (2023) and assume that the search cost $\mathrm{sc}_{ifdm} \geq 0$ is i.i.d. Gumbel-preserving with c.d.f.:

$$G^{\mathrm{sc}}_{ifdm}(z) = \begin{cases} 0 & \text{if } z < 0 \\ \exp\left(-\overline{\mathrm{sc}}^{f}_{idm}\right) & \text{if } z = 0 \\ \dfrac{1 - \exp\left(-\exp\left(-H_0^{-1}(z) - \overline{\mathrm{sc}}^{f}_{idm}\right)\right)}{1 - \exp\left(-\exp\left(-H_0^{-1}(z)\right)\right)} & \text{if } z > 0, \end{cases} \tag{5}$$

with H_0 defined in (2) and $\overline{\mathrm{sc}}^{f}_{idm} \geq 0$. When $\overline{\mathrm{sc}}^{f}_{idm} = 0$, $G^{\mathrm{sc}}_{ifdm}(z) = 1$ for any $z \geq 0$. Given G^{sc}_{ifdm}, we show in Appendix A that $\mathbb{E}[\mathrm{sc}_{ifdm}] = \overline{\mathrm{sc}}^{f}_{idm}$. Proposition 1 in Moraga-González et al. (2023) then shows that this Gumbel-preserving distribution G^{sc}_{ifdm} implies that w_{ifdm} in (3) is i.i.d. Gumbel$(\delta_{ifdm} - \overline{\mathrm{sc}}^{f}_{idm}, 1)$, with δ_{ifdm} the expected maximal indirect utility from $\mathcal{G}_f$ as in (2) and $\overline{\mathrm{sc}}^{f}_{idm} \geq 0$ the expectation of search cost sc_{ifdm}.

We specify $-\overline{\mathrm{sc}}^{f}_{idm} = \gamma_d \,\mathrm{dist}_{fm}$, where $\gamma_d \leq 0$ is a group-specific parameter and dist_{fm} is the average driving distance from municipality m to the closest car dealer of producer f. Given this and the i.i.d. Gumbel assumptions on ϵ_{ijdm} and w_{ifdm}, $\Pr_{idm}(j \mid f)$ and $\Pr_{idm}(f)$ on the right-hand side of (4) are both multinomial logit, and consequently purchase probability $\Pr_{idm}(j)$ simplifies to:

$$\Pr_{dm}(j \mid \nu_i) = \frac{\exp(\delta_{jd} + \mu_{jdm}(\nu_i) + \gamma_d \mathrm{dist}_{f(j),m})}{1 + \sum_{k=1}^{J} \exp(\delta_{kd} + \mu_{kdm}(\nu_i) + \gamma_d \mathrm{dist}_{g(k),m})}, \tag{6}$$

where $f(j)$ denotes the producer f that sells car model j. For each demographic group, we consider two types of consumers: with probability ψ_d consumers are willing to purchase cars online, while with probability $1 - \psi_d$ they are not—even if some producer, such as Tesla, offers this possibility. As discussed in Section 4.5, we compute the probability ψ_d based on questions we asked in an online survey. We distinguish these two types of consumer to better deal with the case of Tesla, which in our sample period was the only producer that offered the possibility to purchase its car models both in person (from one of its car dealers) and online (directly from its website).

People who are not willing to purchase cars online must search for all car models, including Tesla's, by visiting car dealers in person. Conditional on the random coefficients ν_i, the purchase probability of these consumers is (6). In contrast, people willing to purchase cars online have the additional possibility of purchasing any of Tesla's models directly from Tesla's website, potentially without visiting any of Tesla's car dealers in person. For these consumers, we model the potential reduction in average search costs for Tesla (e.g., due to fewer visits to Tesla's car dealers) as $-\overline{\text{sc}}_{idm}^{\text{tesla}} = \tau_d \gamma_d \, \text{dist}_{\text{tesla},m}$, with $\tau_d \in [0,1]$. However, with the exception of this modification, the probability of purchasing car model j for these consumers is analogous to (6). We denote this purchase probability with reduced average search costs for Tesla as $\Pr_{dm}(j \mid \nu_i, \tau_d)$.

As detailed in Section 4.2, we observe car purchases at the level of the municipality m (a town in France) by demographic group d. We obtain purchase probabilities at this level of aggregation by integrating $\psi_d \cdot \Pr_{dm}(j \mid \nu_i, \tau_d) + (1-\psi_d) \cdot \Pr_{dm}(j \mid \nu_i)$ over the distribution F of the random coefficients ν_i:

$$s_{jdm}\left(\delta_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d\right) = \psi_d \cdot \int \Pr_{dm}(j \mid \nu_i, \tau_d) dF(\nu_i) + (1-\psi_d) \cdot \int \Pr_{dm}(j \mid \nu_i) dF(\nu_i), \quad (7)$$

where $\delta_d = (\delta_{1d}, \ldots, \delta_{Jd})$.

As in D'Haultfœuille et al. (2019), we assume that each transaction price p_{jd} (with $p_{jd} = \overline{p}_j$ for any $j \in \mathcal{G}_{\text{tesla}}$ and group d) is chosen at the national level (more details on this below). To obtain the national-level market share of car model j for group d, we average (7) over municipalities:

$$s_{jd}(\delta_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d) = \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}(\delta_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d), \quad (8)$$

where $\mathcal{M}$ collects all municipalities. The relative weight $w_{dm} \equiv M_{dm}/M_d$ measures the incidence of demographic group d in municipality m relative to the country, where M_{dm} and M_d are the observed numbers of consumers of group d in municipality m and throughout the country, respectively, with $\sum_{m \in \mathcal{M}} M_{dm} = M_d$.

2.3 Third-degree price discrimination

We consider a Bertrand-Nash price-setting game in which all producers but Tesla implement third-degree price discrimination by choosing different prices for each demographic group $d = 1, ..., D$. Each producer f (other than Tesla) selects a menu of national prices $p_j = (p_{j1}, ..., p_{jD})$ for each car model they sell by maximizing the national-level profit function

$$\pi_f = \sum_{d=1}^{D} \phi_d \sum_{j \in \mathcal{G}_f} s_{jd}(p_d) \cdot (p_{jd} - c_{jd}), \tag{9}$$

where $p_d = (p_{1d}, ..., p_{Jd})$ is the vector of all prices for group d, $s_{jd}(p_d)$ is the market share defined in (8) (where we highlight its dependence on p_d), and c_{jd} is the marginal cost of car model j for group d. Variable $\phi_d \equiv M_d/M$ is the observed group-specific population weight, where M_d is the national population of group d and M is the national population over all groups. For Tesla, (9) involves fewer prices to be chosen, since $p_{jd} = \overline{p}_j$ for all $j \in \mathcal{G}_{\text{tesla}}$ and $d = 1, ..., D$.

The J first-order conditions associated with demographic group d are

$$p_d = c_d - \widetilde{\mathcal{D}}_d(p_d)^{-1} s_d, \tag{10}$$

where c_d is the vector of group-specific marginal costs of all car models, s_d is the vector of group-specific market shares for all car models, $\widetilde{\mathcal{D}}_d(p_d) = \mathcal{H} \odot \mathcal{D}_d(p_d)$, $\mathcal{H}$ is the ownership matrix, and $\mathcal{D}_d(p_d)$ is the matrix of derivatives of s_d with respect to p_d, with typical element (j, k) equal to $\partial s_{kd}/\partial p_{jd}$.

3 Identification and estimation

3.1 Identification of the unobserved transaction prices

Compared to standard demand models, the identification of $\theta = (\theta_1, ..., \theta_D)$, where $\theta_d \equiv (\beta_d, \alpha_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d)$, presents the additional complication that the group-specific transaction prices $p(\theta) = (p_1(\theta), ..., p_D(\theta))$ are not observed by the researcher (with the exception of Tesla's transaction prices, which equal the observed list prices). Following D'Haultfœuille et al. (2019), we address this complication by relying on both demand and supply restrictions to jointly identify preference parameters θ and transaction prices

$p(\theta)$.[7]

We make the following assumptions, which are sufficient for the identification of the unobserved transaction prices (see details in D'Haultfœuille et al., 2019).

A1. Observability of national-level group-specific market share s_{jd} for each j and d.

A2. Constant marginal costs across demographic groups, $c_{jd} = c_j$ for any j and d.

A3. Relevance of list prices. For each car model j, we assume that the list price $\overline{p}_j$ satisfies $\overline{p}_j = \max\{p_{j1}, ..., p_{jD}\}$; that is, the list price is the highest transaction price paid by at least one demographic group.

Intuitively, assumptions A1-A3 allow us to back out, for any given value of the demand parameters θ, the transaction prices $p(\theta)$ that rationalize both demand and supply. With these, we are back to a standard model in which all prices are observed.

First, given assumption A1 and following Berry (1994), for a given group d and some value of $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, we obtain $\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d) = (\delta_{jd}(\Pi_d, \Sigma_d, \gamma_d, \tau_d))_{j=1,...,J}$ by inverting the system of J national-level market share equations given by (8). Second, we obtain the transaction prices corresponding to θ. To this end, note that the first-order conditions (10) and assumptions A2-A3 imply

$$\overline{p}_j = c_j - \min\left\{\left[\widetilde{\mathcal{D}}_1{}^{-1}s_1\right]_j, ..., \left[\widetilde{\mathcal{D}}_D{}^{-1}s_D\right]_j\right\}, \tag{11}$$

with $[x]_j$ denoting the j-th element of vector x. In the absence of a random coefficient on price, the matrices $\widetilde{\mathcal{D}}_1, ..., \widetilde{\mathcal{D}}_D$ do not depend on transaction prices: they only depend on $\varsigma \equiv (\alpha_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d)_{d=1,...,D}$ through $(\alpha_d, \delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d))_{d=1,...,D}$. Then, for a given value of ς, each transaction price can be obtained as

$$p_{jd}(\varsigma) = \overline{p}_j + \min\left\{\left[\widetilde{\mathcal{D}}_1^{-1}s_1\right]_j, ..., \left[\widetilde{\mathcal{D}}_D^{-1}s_D\right]_j\right\} - \left[\widetilde{\mathcal{D}}_d^{-1}s_d\right]_j, \tag{12}$$

where we leave the dependence of $\widetilde{\mathcal{D}}_d$ on ς implicit.

Assumption A1 relates to data availability and is the main reason why p_{jd} and ξ_{jd} vary at the level of (j, d) rather than at the more disaggregate level of (j, d, q), where q denotes a specific car dealer. Handling unobserved transaction prices at this level would require precise measures of purchase probabilities at the (j, d, q) level, which are not

[7]Note that, even if transaction prices were observed, without further "extrapolating" assumptions, the prices that consumers face for the car models they do *not* purchase remain unobserved and a similar procedure would still be needed.

generally available. Even with q-specific purchase data, the limited number of sales at any individual car dealer would lead to imprecisely measured purchase probabilities, with severe consequences in terms of measurement error that cannot be easily addressed in nonlinear structural models (see Freyberger, 2015; Gandhi et al., 2019, and the discussion below).

3.2 Moment Conditions

Demand-side moments. We compute the empirical counterpart of the following moment conditions

$$\mathbb{E}\left[Z'_{jd}\xi_{jd}\right] = 0, \quad d = 1, ..., D, \tag{13}$$

with Z_{jd} a group-specific vector of instruments. To do this, we first compute $\xi_{jd}(\beta_d, \varsigma) = \delta_{jd}(\Pi_d, \Sigma_d, \gamma_d, \tau_d) - X'_j\beta_d - \alpha_d p_{jd}(\varsigma)$ and then consider the empirical moment condition $g_1(\theta) = (g_{11}(\beta_1, \varsigma), ..., g_{1D}(\beta_D, \varsigma))$, where

$$g_{1d}(\beta_d, \varsigma) = \frac{1}{J}\sum_{j=1}^{J} Z'_{jd}\xi_{jd}(\beta_d, \varsigma) = 0. \tag{14}$$

As usual, while transaction prices are endogenous by construction, we assume the observed characteristics X_j to be exogenous. Valid instruments can be obtained as functions of the exogenous characteristics of car model j and those of other car models.

Supply-side moments. We assume that the natural logarithm of marginal cost c_j is a linear combination of observed car characteristics X_j, cost shifters W_j, and an unobserved cost shock ω_j, such that

$$\ln(c_j) = X'_j\lambda_1 + W'_j\lambda_2 + \omega_j, \tag{15}$$

where we assume that (X_j, W_j) are uncorrelated with ω_j and $(\xi_{jd})_{d=1,...,D}$. We compute the empirical counterpart of the following moment conditions

$$\mathbb{E}\left[Z'_{jS}\omega_j\right] = 0, \tag{16}$$

with Z_{jS} a vector of supply-side instruments. The associated supply-side moment conditions are

$$g_2(\varsigma, \lambda) = \frac{1}{J}\sum_{j=1}^{J} Z'_{jS}\omega_j(\varsigma, \lambda) = 0, \tag{17}$$

where $\lambda = (\lambda_1, \lambda_2)$ and $\omega_j(\varsigma, \lambda)$ can be computed using (11) and (15). Again, valid instruments Z_{jS} can be obtained as functions of (X_j, W_j) and of (X_k, W_k), $k \neq j$.

Micro moments. We complement moment conditions (13) and (16) with micro moments that help identify the nonlinear parameters $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$. To do this, we take advantage of the fact that we observe purchase probabilities at the demographic group-by-municipality level, s_{jdm}, and of the additional consumer-level survey data we collected. We illustrate here the idea of these micro moments for the variable dist_{fm} and follow a similar procedure for the demographics dem_{dm} and their interactions with X_j. The full list of micro moments is then presented in Section 5.

We construct micro moments for dist_{fm} by matching the observed and predicted average distance between consumers and the car dealers of the purchased car models. In particular, we specify $g_3(\Pi, \Sigma, \gamma, \tau) = (g_{31}(\Pi_1, \Sigma_1, \gamma_1, \tau_1), ..., g_{3D}(\Pi_D, \Sigma_D, \gamma_D, \tau_D))$, where each $g_{3d}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ contains two types of micro moments, the first helpful to identify the distance parameter γ_d (common across all producers) and the second to identify the Tesla-specific potential reduction in average search costs τ_d:

$$\underbrace{\frac{\sum_{j=1}^{J} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d) \cdot \text{dist}_{f(j),m}}{\sum_{j=1}^{J} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)}}_{\text{Predicted distance overall: } \overline{D}^{\text{all}}(\gamma_d, \tau_d)} - \frac{\sum_{j=1}^{J} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm} \cdot \text{dist}_{f(j),m}}{\sum_{j=1}^{J} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}},$$

$$\underbrace{\frac{\sum_{j \in \mathcal{G}_{\text{tesla}}} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d) \cdot \text{dist}_{\text{tesla},m}}{\sum_{j \in \mathcal{G}_{\text{tesla}}} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)}}_{\text{Predicted distance for Tesla: } \overline{D}^{\text{tesla}}(\gamma_d, \tau_d)} - \frac{\sum_{j \in \mathcal{G}_{\text{tesla}}} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm} \cdot \text{dist}_{\text{tesla},m}}{\sum_{j \in \mathcal{G}_{\text{tesla}}} \sum_{m \in \mathcal{M}} w_{dm} \cdot s_{jdm}}, \tag{18}$$

where s_{jdm} is the observed purchase probability of group d for car model j in municipality m. The corresponding purchase probability predicted by the model is instead obtained by evaluating (7) at any given $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$: $s_{jdm}(\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d), \Pi_d, \Sigma_d, \gamma_d, \tau_d) = s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, where we obtain $\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ by inverting the system of J national-level market shares in (8). Note that for any given $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, we have $s_{jd} = s_{jd}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ and the observed denominators in (18) are equal to their predicted counterparts. However, only at the true value of $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, say $(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0)$, do we also have the guarantee that $s_{jdm} = s_{jdm}(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0)$.

Three points are worth noting. First, if we only have a few moment conditions in (13) and (16), we may not be able to identify $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$. In these cases, additional moments, such as the micro moments we propose, become necessary to identify $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$.

Moreover, the instruments in (13) and (16) may lack the power to estimate with sufficient precision the distance parameter γ_d and the Tesla-specific potential reduction in average search costs τ_d (for example). Instead, we expect the micro moments in (18) to be particularly informative about γ_d and τ_d.

Intuitively, as γ_d becomes more negative, car models with more distant dealers suffer a larger loss relative to those with closer dealers, and the outside option becomes more attractive in municipalities where dealers are far. Both channels shift the composition of purchases toward closer dealers, so $\overline{D}^{\text{all}}(\gamma_d, \tau_d)$ decreases with more negative γ_d. Since Tesla represents a small share of total purchases, $\overline{D}^{\text{all}}(\gamma_d, \tau_d)$ is approximately invariant to τ_d, making the first micro moment particularly informative about γ_d. Similarly, $\overline{D}^{\text{tesla}}(\gamma_d, \tau_d)$ also decreases as γ_d becomes more negative. Additionally, a higher τ_d does not greatly affect the relative attractiveness of Tesla within any given municipality (since all Tesla products share the same $\text{dist}_{\text{tesla},m}$), but it does shift the composition of Tesla buyers toward municipalities closer to Tesla dealers, reducing $\overline{D}^{\text{tesla}}(\gamma_d, \tau_d)$. Therefore, the second micro moment is expected to be particularly useful for identifying τ_d.

Second, these micro moments remain valid even if dist_{fm} is endogenous. The observed distances could be correlated with $(\xi_{jd})_{j,d}$ if, for example, producers partially or fully observed $(\xi_{jd})_{j,d}$ at the time of deciding where to locate their car dealers. As mentioned above, at the true value of the parameters $(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0)$, $\delta_{jd}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ will equal its true value $\delta_{jd}(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0) = \delta_{jd}^0$, and thus $s_{jdm}(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0) = s_{jdm}^0$. As this only follows from Berry (1994)'s demand inverse, it will hold regardless of any dependence between the observed distances and the unobserved components of demand $(\xi_{jd})_{j,d}$.

Third, assuming that s_{jd} is measured without error is reasonable (and standard) given the large number of consumers in each demographic group throughout the country. In contrast, assuming that the municipal-level purchase probability s_{jdm} is also measured without error may be strong. We only observe a proportion computed from a finite sample instead of the true purchase probability (say, s_{jdm}^0), and the corresponding sample is small for small municipalities. However, regardless of measurement error in s_{jdm}, we have

$$\mathbb{E}\left[\sum_{j=1}^{J}\sum_{m\in\mathcal{M}} w_{dm}\cdot s_{jdm}\cdot \text{dist}_{f(j),m} \;\Big|\; \left(\text{dist}_{f(j),m}\right)_{j,m}, (\xi_{jd})_{j,d}\right] = \sum_{j=1}^{J}\sum_{m\in\mathcal{M}} w_{dm}\cdot s_{jdm}^0\cdot \text{dist}_{f(j),m}.$$

That is, even if s_{jdm} is measured with error (but maintaining, as usual, that the $(s_{jd})_{j=1,\ldots,J}$ are measured without error), at the true value of the nonlinear parameters $(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0)$, we still obtain $\mathbb{E}[g_{3d}(\Pi_d^0, \Sigma_d^0, \gamma_d^0, \tau_d^0)] = 0$. In other words, the micro moments in (18) are robust to this form of measurement error. For the same reason, with these micro moments, the presence of "zeros" in the observed municipal-level purchase probabilities does not raise any particular concern (Gandhi et al., 2019).

3.3 Two-step GMM estimation

Although (θ, λ) could be simultaneously estimated by the generalized method of moments (GMM) using all moments $g(\theta, \lambda) = (g_1(\theta),\ g_2(\varsigma, \lambda),\ g_3(\Pi, \Sigma, \gamma, \tau))$, this would be computationally intensive as it requires repeatedly solving both demand inverses and the system of equations (12) (D'Haultfœuille et al., 2019). Given the absence of a random coefficient on price in (1), we can simplify implementation by estimating (θ, λ) in two sequential GMM steps that separate these two tasks.[8]

First, we use the micro moments $g_3(\Pi, \Sigma, \gamma, \tau)$ to estimate the nonlinear parameters $(\Pi, \Sigma, \gamma, \tau)$. Second, given $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$, we estimate the other demand parameters $(\beta, \alpha) = (\beta_d, \alpha_d)_{d=1,\dots,D}$ and the marginal cost parameters $\lambda = (\lambda_1, \lambda_2)$ using the remaining moment conditions $(g_1(\beta, \alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}),\ g_2(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}, \lambda))$. Importantly, because the average group-specific utilities $\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ are fully determined by $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, this second step takes them as given.

As each step is standard, we report all computational details in Appendix C.2. To account for the estimation error in $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$, we compute the variance-covariance matrix of the second-step estimator using the formulae in Newey and McFadden (1994).

4 Data

4.1 Demographic data

We assemble a dataset of municipal-level demographics from the National Institute of Statistics and Economic Studies (INSEE) to characterize the set of potential car buyers.[9] Metropolitan France comprises 35,296 municipalities with an average population of roughly 1,350 inhabitants. Our data include yearly population censuses, median income by age category, household size, and population density.

We divide consumers into three age categories (below 40, 40 to 59, and above 60) and

[8] As noted in footnote 6, more general specifications with a random coefficient on price (estimated by a one-step GMM) yielded qualitatively similar results, and we opted for the simpler specification given its substantially lower computational complexity.

[9] Paris, Marseille, and Lyon, the three largest cities, are further split into *arrondissements*, which we treat as separate municipalities throughout.

two income categories (low and high income) to form six demographic groups. These groups are easily observable by car dealers and potentially associated with heterogeneous preferences, thus forming a basis for third-degree price discrimination. We do not observe consumer income directly in the car registration data, so we assign each consumer an income category based on age and municipality of residence. For each age category, we divide municipalities into low and high income based on median income at the municipal-by-age category level. As a consequence, consumers of the same age and in the same municipality are assigned to the same demographic group. However, different age categories within a municipality can be allocated to different income categories.

We summarize these data at the level of our demographic groups in Appendix Table A.1 and we provide additional details on the construction of the dataset in Appendix Appendix C.

4.2 Distance and sales data

Distance to car dealers. We collect a novel dataset of car dealer locations in France directly from manufacturers' websites. The data include the name, address, and brands sold for each car dealer, covering 7,241 brand-dealer combinations. We convert addresses to geographic coordinates to compute driving distances.[10]

We use OpenStreetMap for the coordinates and footprint areas of buildings in France to compute average driving distances from each municipality to the car dealers of each brand.[11] For each municipality m, we randomly select 100 buildings and calculate the driving distance from each building to every car dealer. We then compute the average driving distance from m to each car dealer of brand f, weighting by building area. This yields a set of average driving distances from m to all car dealers of brand f: $\text{dist}^1_{fm}, \text{dist}^2_{fm}, \text{dist}^3_{fm}, \ldots$ Finally, we set $\text{dist}_{fm} = \min\{\text{dist}^1_{fm}, \text{dist}^2_{fm}, \text{dist}^3_{fm}, \ldots\}$, the average driving distance from m to the closest car dealer of brand f.

While straight-line distances between buildings and car dealers are simple to compute, driving distances, which account for the road network and speed limits, are more relevant for consumers' search costs but costly to obtain for all building-dealer pairs. We therefore proceed in two steps. First, we construct a training sample of 969,455 municipal-zipcode pairs for which we observe both the straight-line distance (between centroids) and the

[10] The data were collected in 2024, as reliable historical data on dealer locations are not readily available for France.

[11] We exclude all buildings with a footprint area below 25 sq. meters or above 500 sq. meters because they are likely not residential buildings.

driving distance obtained from TomTom's API.[12] Second, we estimate a predictive model of driving distances as a function of straight-line distances using fractional polynomial regressions, and apply it to predict driving distances for all building-dealer pairs.

Car registrations. We obtain proprietary data on all new car registrations in France from 2009 to 2021 from AAAData.[13] The data are aggregated at the municipal level and by age group (in increments of 5 years).

Sales are recorded at the level of the brand (29 brands), model (372 models), engine type (gasoline, diesel, electric, plug-in, hybrid), and body trim (sedan, convertible, station wagon). The data include common car attributes such as horsepower, weight, CO_2 emissions, and fuel consumption, as well as the list price.[14] List prices are adjusted to be net of the French feebate program.[15] We complement the dataset with annual average fuel and electricity prices to construct a measure of driving cost (in euros per 100km). Both list prices and driving costs are deflated to 2018 euros. We obtain the market segment (e.g., subcompact, compact, SUV, etc.) of each car model from additional proprietary data provided by Jato Dynamics.[16]

We define a product as a combination of a brand, a model, an engine type, and a body trim. Whenever multiple versions of the same product exist (e.g., different trim levels), we keep the characteristics of the most frequently purchased one. After aggregating sales by product, demographic group, and year, the final dataset includes 4,960 observations over 13 years.

Descriptive statistics. We present descriptive statistics on car sales, distances to car dealers, and car characteristics in Appendix Table A.5. Panel A breaks down sales by product and demographic group. Groups 4 and 6, representing high-income consumers aged 40 or older, purchase on average more than twice as many vehicles as other groups.

Panel B reports driving distances from consumers to the closest dealer of each brand, weighted by brand-level sales and group-specific municipal populations to preserve comparability across groups. Consumers in groups 3 and 5 (low income, aged 40 or older)

[12]See https://developer.tomtom.com/.

[13]Source: https://www.aaa-data.fr/. AAAData collects and cleans the central vehicle registration database (SIV) and assigns a price based on manufacturers' catalogs.

[14]We encounter a few missing observations on key car characteristics (horsepower is missing for electric vehicles). In these cases, we fill the missing values with registration data from SIV.

[15]The French feebate program offers incentives to promote low-emission vehicles, based on engine type and tailpipe emissions.

[16]Source: https://www.jato.com.

live significantly further away from car dealers than other groups. Car dealers are also on average closer to high-income than to low-income municipalities.

Panel C summarizes the car characteristics. Since all consumers face the same set of products, we report a common set of statistics weighted by total sales.

4.3 Evidence of price dispersion

In this section, we provide evidence that income and age are the most relevant observable demographics that correlate with price dispersion in the French car industry.

We use two waves (2011 and 2017) of a French survey of consumers' expenditures.[17] The survey records demographic characteristics of respondents, as well as some vehicle characteristics (model, engine type; i.e., gasoline, diesel, electric, or hybrid) and the transaction prices of car purchases made in the year preceding the survey. For each car purchase, we observe whether the vehicle was new or second hand and whether it was purchased from a car dealer or through a private sale; we restrict the sample to dealer purchases.[18] When applicable, the trade-in value of the consumer's old vehicle is also recorded. We regress transaction prices on a set of consumer demographics and fixed effects. We consider transaction prices that include and exclude trade-in values. We also consider specifications that restrict the sample to new vehicles only. To be consistent with our structural model, we measure income as the median income of the consumer's age group in their municipality of residence rather than using individual income. All specifications control for car model $\times$ engine type $\times$ new/used and year $\times$ month fixed effects. The results are presented in Appendix Table A.2.

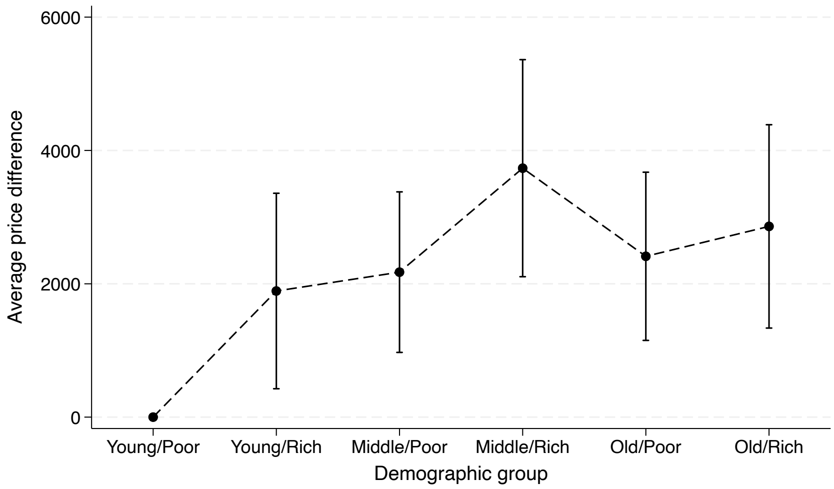
Our estimates indicate that high-income consumers pay more on average for the same vehicle. Our results also suggest that age correlates positively with transaction prices, although the effect is not statistically significant when restricting to new vehicles. We do not find a significant correlation for other demographics observable by car dealers, such as gender, household size, or urbanity.[19] The absence of a gender effect is unsurprising, as car purchases are typically household-level decisions and the survey respondent need not be the person who negotiated the price.

[17] Source: "Enquête Budget des Familles (BDF) - 2011 and 2017 (INSEE)", accessed through CASD (DOI: http://datapresentation.casd.eu/10.34724/CASD.12.1020.V4 and https://doi.org/10.34724/CASD.12.3581.V1).

[18] We also exclude observations in which the vehicle was purchased following an insurance claim (i.e., the replacement of a damaged vehicle).

[19] Some specific household sizes correlate with price dispersion (four and five members); however, these are unlikely to be observable by car dealers. All regressions include indicator variables for the buyer's country of origin (not reported in Appendix Table A.2), which are also mostly insignificant.

Figure 1: Evidence of price dispersion among demographic groups

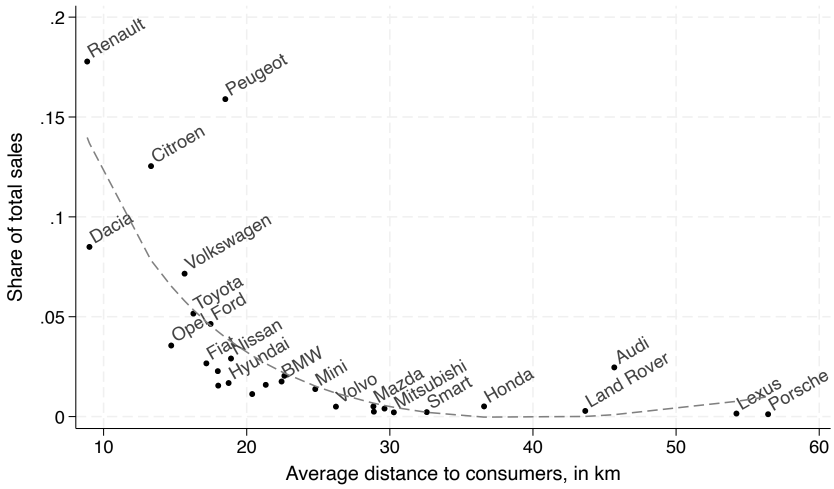


Notes: This figure presents the estimated coefficients from a regression of transaction prices on demographic group indicators, based on a survey of consumers' expenditures (see Table A.3, column 1). The regression controls for gender, household size, urbanity, and buyer country of origin, and includes car model × engine × new/used and year × month fixed effects. The bars represent 95% confidence intervals, based on standard errors clustered at the car model × engine × new/used level.

The correlation between income and prices may partly reflect the choice of unobserved options, if wealthier consumers tend to select more expensive configurations within a given model. However, the estimated coefficient on income is smaller when we restrict the sample to new vehicles (columns (2) and (4)), where the scope for option customization is larger than for second-hand vehicles. This pattern suggests that unobserved options are unlikely to fully explain the observed price dispersion.

To quantify the magnitude of price differences across our demographic groups, we re-estimate the regression replacing income and age with indicators for each of our six groups. Again, we control for gender, household size, urbanity, and the same fixed effects. Figure 1 shows the estimated coefficients associated with these demographic group indicators, while the details of the regression results are reported in Appendix Table A.3. Middle-aged high-income consumers face the highest transaction prices, paying roughly €3,700 more than young low-income consumers, the baseline group. Old high-income consumers follow at roughly €2,900 more. Overall, the estimated price differences range from €1,900 to €3,700.

Figure 2: Market share by brand and dealer proximity



Notes: Market shares are computed as the ratio of each brand's total sales to total sales across all brands from 2009 to 2021. Distance is measured as the average driving distance to consumers over all demographic groups. The dashed line represents fitted values from a fractional polynomial regression. Tesla is excluded from the figure; its average distance to consumers is 96 km.

4.4 Evidence of search costs

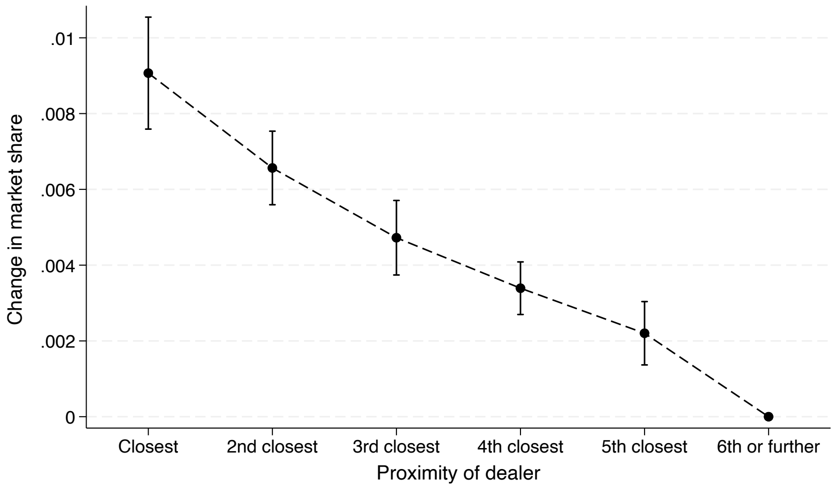
We now show that distance to car dealers, our proxy for search costs, correlates with consumer purchasing patterns.

Figure 2 plots the aggregate market share of each brand against the average driving distance to consumers. Brands with dealers located closer to consumers tend to capture a larger market share.[20] Additional statistics on distances and market presence by brand are reported in Appendix Table A.4.

We also estimate a regression of brand-level market shares at the municipal level on indicators for the proximity rank of each brand's nearest dealer (closest, second closest, and so on), controlling for municipality and brand × year fixed effects. Figure 3 plots the estimated coefficients and shows that market shares decrease monotonically with proximity rank. For instance, Renault's market share is larger in municipalities where it is the closest dealer than where it is the second closest, or third closest.

[20] A similar pattern emerges when plotting market shares against the number of dealers per brand.

Figure 3: Market share advantage from car dealer proximity



Notes: This figure presents estimated coefficients from a regression of brand-level market shares on proximity rank indicators (closest dealer, second closest, etc.). Market shares are computed within the car market (excluding the outside option), pooling sales across products and demographic groups within each brand. The regression controls for municipality and brand $\times$ year fixed effects. The bars represent 95% confidence intervals, based on standard errors clustered at the municipality level.

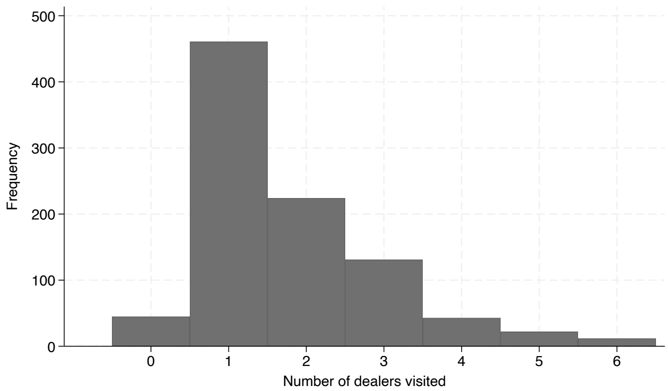
4.5 Survey evidence on search and propensity to shop online

We complement our analysis with data from an online survey of recent car buyers, conducted in November 2025 through Qualtrics. The survey targets a representative panel of French residents, stratified by age (six categories), gender, and region of residence (thirteen regions). It covers 2,258 respondents who purchased a car in the last ten years and collects information on the vehicle purchased, search behavior, use of online resources, and willingness to purchase online. Since respondents were allowed to skip questions, effective sample sizes vary across questions.

Respondents visited on average 1.8 dealers before purchasing their car (1.7 for new car buyers). Figure 4 shows the full distribution: a substantial share of consumers visited only one dealer, while very few visited more than three. Overall, 10.8% of respondents purchased their car online, a figure that likely reflects the prevalence of online marketplaces for used vehicles. For new cars, this share drops to 5.8%, confirming that online new car sales remain limited. However, 67% consulted online resources to learn about car models before purchasing, and 81% intend to do so before their next purchase, suggesting that consumers are already accustomed to using online channels during the search process.

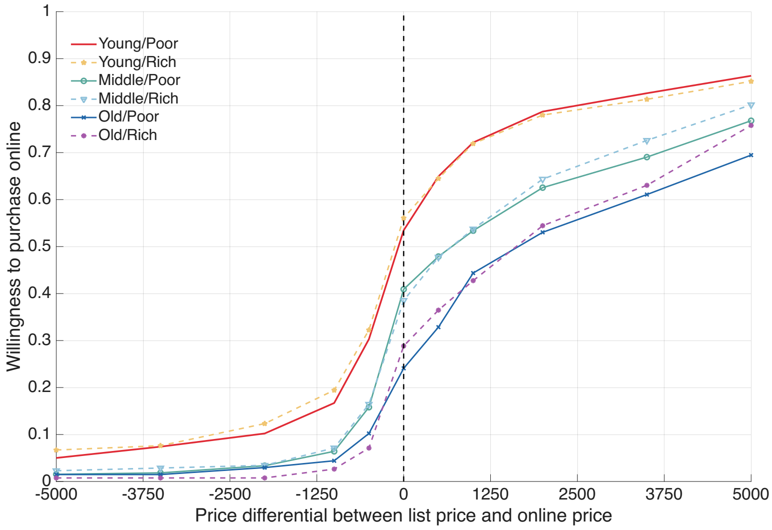
We also elicit respondents' willingness to purchase a car online at various price levels

Figure 4: Distribution of the number of dealers visited



Notes: This figure illustrates the number of dealers visited. We denote by "0" the consumers who purchased a car online.

Figure 5: Willingness to purchase online as a function of the price differential



Notes: This figure illustrates the share of consumers willing to make an online purchase as a function of the price differential between the list price and a hypothetical online price. Positive values indicate that the consumers are compensated to shop online. Negative values indicate that consumers must pay a surcharge to shop online.

using an iterative bidding procedure. We first ask whether the respondent would be willing to buy online at the same price as in person. If yes, we offer successive surcharges until the respondent declines. If no, we offer successive discounts until the respondent accepts.[21] Figure 5 displays the resulting cumulative distribution of willingness to purchase online as a function of the price differential, separately for each demographic group. The curves are roughly parallel, suggesting similar within-group heterogeneity but different baseline propensities to purchase online.

[21] The proposed surcharges and discounts are €500, €1,000, €2,000, €3,500, and €5,000.

In our counterfactual analysis, we distinguish between consumers who would consider purchasing online and those who are captive to the in-person channel. We define the former as respondents who would be willing to purchase online at the same price as in person, corresponding to the share at zero price differential in Figure 5. Table 1 reports these shares by demographic group. Since these shares are well below one, a substantial fraction of consumers within each group remains captive to the in-person channel, motivating the distinction between captive and non-captive consumers in our model. The propensity to consider online purchasing decreases sharply with age, from 53–58% for young consumers to 24–29% for old consumers. Income plays a limited role within age groups.

Table 1: Share of consumers willing to purchase cars online

Group	Share	Observations
Group 1: Young/Poor	0.53	467
Group 2: Young/Rich	0.58	213
Group 3: Middle/Poor	0.41	331
Group 4: Middle/Rich	0.39	356
Group 5: Old/Poor	0.24	137
Group 6: Old/Rich	0.29	267
All consumers	0.43	1,771

Notes: Share of respondents willing to purchase a car online at the same price as in person, corresponding to the value of the cumulative distribution at zero price differential in Figure 5.

5 Estimation results

We define products as brand, model, engine type, and body trim combinations, treat each year as a separate market, and set the potential market for each demographic group to one-quarter of the number of households in that group per year. The demand specification includes the list price (net of the French feebate program, in €10,000), horsepower (in 100kW), weight (in 1,000kg), fuel cost (in euros per 100km), engine type and body trim fixed effects, the driving distance to the closest car dealer of each brand (in 10km), and brand and year fixed effects. The vector of characteristics X_j includes a constant, common to all $j \neq 0$, interacted with $\beta_d^{\text{const}} + \Pi_d^{\text{const}} \text{dem}_{dm} + \Sigma_d \nu_i$, where dem_{dm} includes average income, average household size, and an urban indicator, and ν_i is a scalar normally distributed random coefficient with standard deviation Σ_d.[22] This

[22] We experimented with the inclusion of additional random coefficients interacted with other elements of X_j, but did not obtain any significant estimates for the corresponding Σ_d. We therefore present our results for this more parsimonious specification that only includes a random coefficient on the constant.

flexible intercept allows consumers to have different outside options, which is crucial to avoid overstating the market expansion induced by the online distribution channel.

The marginal cost specification includes horsepower, weight, fuel consumption (in liters per 100km), engine type and body trim fixed effects, and a time trend. We include two cost shifters: an input price index, computed by interacting the average yearly price of key inputs (steel, plastics, and aluminum, assumed to constitute 77%, 12%, and 11% of a car's weight, respectively) with the car's weight; and the real exchange rate interacted with the car's country of origin, following Grieco et al. (2023).[23] Both cost shifters are lagged by one year to reflect planning horizons.

We address price endogeneity using both demand-side and supply-side instruments. The demand-side instruments Z_j are common across demographic groups ($Z_{jd} = Z_j$ for all d) and include, in addition to X_j, sums of competitors' exogenous characteristics (horsepower, weight, fuel cost) and counts of competing products (overall, with the same engine type, and with the same body trim). The supply-side instruments Z_{jS} follow a similar structure, using horsepower, weight, fuel consumption, both cost shifters, and analogous counts of competing products.

As discussed in Section 3, our estimator is robust to potential endogeneity of the observed distances without requiring additional instruments. More details on the empirical specification and each estimation step can be found in Appendix C.2. We present the estimates of the first-step GMM (the nonlinear parameters) in Table 2 and of the second-step GMM (the linear parameters) in Table 3.

The third panel of Table 2 summarizes the 858 micro moments used for the estimation of the nonlinear parameters $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)_{d=1,\ldots,6}$ (11 "types" of micro moments × 6 demographic groups × 13 years). The first two panels of Table 2 show that the distance coefficients and those of the observed demographics are precisely estimated and have the expected signs. All demographic groups dislike traveling farther away to purchase cars, with middle-aged and older consumers facing the highest average search costs. The estimated average reduction in search costs for Tesla's online sales is $100 \cdot (1 - \sum_d \phi_d \widehat{\tau}_d) = 37\%$. Wealthier and larger households are more likely to purchase a car, while most consumers living in urban municipalities are less likely to own one (the exception being the young/poor group), perhaps because of the (relative) lack of parking and the availability of better public transport.

[23] The real exchange rate captures differences in the cost of labor for each brand and is taken from Penn World Tables 10.0, `pl_con`. See Feenstra et al. (2015).

Table 2: Estimates, nonlinear parameters (first-step GMM)

	Demographic group					
	Young/Poor	Young/Rich	Middle/Poor	Middle/Rich	Old/Poor	Old/Rich
Nonlinear parameters γ_d and τ_d						
Distance (γ_d)	-0.035	-0.017	-0.059	-0.050	-0.051	-0.052
	(0.004)	(0.005)	(0.004)	(0.005)	(0.004)	(0.006)
Tesla search cost reduction (τ_d)	0.630	0.630	0.630	0.630	0.630	0.630
	(0.242)	(0.242)	(0.242)	(0.242)	(0.242)	(0.242)
Nonlinear parameters Π_d						
Constant × Income	-3.440	0.264	0.770	0.271	1.221	0.334
	(0.466)	(0.051)	(0.161)	(0.030)	(0.219)	(0.043)
Constant × Household size	0.616	0.987	0.494	1.105	0.465	0.821
	(0.173)	(0.028)	(0.024)	(0.032)	(0.028)	(0.030)
Constant × Urban	0.428	-0.577	-0.405	-0.413	-0.556	-0.536
	(0.021)	(0.019)	(0.028)	(0.021)	(0.025)	(0.021)
Horsepower × Income	-0.481	0.955	0.445	0.526	1.677	0.417
	(0.029)	(0.287)	(0.766)	(0.175)	(0.914)	(0.224)
Horsepower × Urban	0.520	0.757	0.580	0.378	0.552	0.536
	(0.759)	(0.083)	(0.132)	(0.097)	(0.115)	(0.104)
Weight × Urban	0.669	-0.963	-0.774	-0.710	-0.911	-0.959
	(0.133)	(0.075)	(0.075)	(0.072)	(0.094)	(0.081)
Micro moments						
$\mathbb{E}(\text{Distance} \mid j \neq 0)$	1.600	1.442	1.823	1.456	2.138	1.328
	[1.599]	[1.441]	[1.823]	[1.455]	[2.137]	[1.329]
$\mathbb{E}(\text{Income} \mid j \neq 0)$	1.703	2.458	1.810	2.547	1.893	2.491
	[1.703]	[2.458]	[1.810]	[2.546]	[1.893]	[2.491]
$\mathbb{E}(\text{Household size} \mid j \neq 0)$	2.249	2.430	2.253	2.394	2.258	2.296
	[2.249]	[2.429]	[2.253]	[2.394]	[2.258]	[2.297]
$\mathbb{E}(\text{Urban} \mid j \neq 0)$	0.379	0.312	0.315	0.279	0.160	0.330
	[0.379]	[0.312]	[0.315]	[0.278]	[0.160]	[0.329]
$\mathbb{C}(\text{Horsepower}, \text{Income} \mid j \neq 0)$	1.266	1.884	1.386	2.001	1.419	1.911
	[1.265]	[1.881]	[1.386]	[1.997]	[1.419]	[1.910]
$\mathbb{C}(\text{Horsepower}, \text{Urban} \mid j \neq 0)$	0.279	0.240	0.241	0.217	0.119	0.250
	[0.279]	[0.239]	[0.241]	[0.217]	[0.119]	[0.249]
$\mathbb{C}(\text{Weight}, \text{Urban} \mid j \neq 0)$	0.644	0.538	0.545	0.486	0.272	0.563
	[0.644]	[0.538]	[0.544]	[0.485]	[0.272]	[0.562]
$\mathbb{E}(\text{Distance} \mid j \in \mathcal{G}_{\text{tesla}})$	5.407	3.887	5.734	3.821	6.543	4.347
	[5.587]	[2.807]	[5.313]	[3.191]	[6.834]	[3.631]
$\mathbb{E}(\text{Income} \mid j \in \mathcal{G}_{\text{tesla}})$	1.080	1.603	1.135	1.675	1.201	1.613
	[1.094]	[1.710]	[1.135]	[1.819]	[1.200]	[1.759]
$\mathbb{E}(\text{Household size} \mid j \in \mathcal{G}_{\text{tesla}})$	1.396	1.491	1.393	1.481	1.393	1.429
	[1.368]	[1.471]	[1.372]	[1.450]	[1.390]	[1.403]
$\mathbb{E}(\text{Urban} \mid j \in \mathcal{G}_{\text{tesla}})$	0.198	0.187	0.191	0.167	0.090	0.175
	[0.268]	[0.261]	[0.296]	[0.257]	[0.100]	[0.264]
Number of micro moments	858					
Value of 1st stage obj. function	0.0671					

Notes: Distance is in 10km, income in €10,000, horsepower in 100kW, weight in 1,000kg. The micro moments used in estimation are at the demographic group × year level. We report the micro moments implied by the model averaged over markets, with their observed counterparts in square brackets.

Table 3: Estimates, linear parameters (second-step GMM)

	Demographic group						Cost function
	Young/Poor	Young/Rich	Middle/Poor	Middle/Rich	Old/Poor	Old/Rich	$\ln(c_j)$
Linear parameters α_d							
Price (α_d)	-3.440	-3.194	-2.845	-2.443	-2.886	-2.021	
	(0.466)	(0.444)	(0.438)	(0.406)	(0.453)	(0.393)	
Linear parameters β_d, λ_1, *and* λ_2							
Constant	-12.117	-11.076	-11.327	-10.343	-10.380	-10.012	-0.929
	(0.649)	(0.601)	(0.578)	(0.510)	(0.637)	(0.516)	(0.087)
Horsepower	5.505	3.425	4.272	2.836	2.010	1.958	0.579
	(0.309)	(0.290)	(0.264)	(0.231)	(0.286)	(0.219)	(0.035)
Weight	3.981	3.950	3.225	2.734	3.325	2.217	0.647
	(0.317)	(0.297)	(0.274)	(0.244)	(0.308)	(0.248)	(0.038)
Fuel cost	-0.264	-0.265	-0.169	-0.150	-0.226	-0.161	
	(0.041)	(0.039)	(0.035)	(0.032)	(0.037)	(0.031)	
Fuel consumption							0.015
							(0.010)
Diesel	-1.052	-0.956	-0.870	-0.796	0.004	0.657	-0.368
	(0.494)	(0.449)	(0.413)	(0.352)	(0.434)	(0.330)	(0.068)
Electric	-0.628	-0.861	-0.275	-0.457	0.057	0.360	-0.216
	(0.440)	(0.394)	(0.363)	(0.302)	(0.384)	(0.284)	(0.058)
Plug-in	-0.390	-0.417	-0.102	-0.241	0.639	1.004	-0.205
	(0.498)	(0.451)	(0.419)	(0.357)	(0.440)	(0.338)	(0.065)
Hybrid	-1.426	-1.488	-0.815	-0.742	-0.449	0.234	-0.204
	(0.552)	(0.504)	(0.470)	(0.398)	(0.480)	(0.362)	(0.069)
Convertible	-0.138	-0.210	-0.018	-0.082	-0.543	-0.763	0.224
	(0.247)	(0.246)	(0.212)	(0.194)	(0.236)	(0.189)	(0.028)
Wagon	-0.062	-0.150	0.028	-0.080	0.032	-0.050	-0.036
	(0.131)	(0.125)	(0.123)	(0.115)	(0.137)	(0.120)	(0.012)
Input price index							-0.214
							(0.100)
Real exchange rate							0.197
							(0.039)
Trend							0.018
							(0.002)
Willingness-to-pay (γ_d/α_d)	10.09	5.28	20.86	20.55	17.77	25.93	
	(1.85)	(1.76)	(3.53)	(4.09)	(3.169)	(5.77)	
Observations	4,960						
Value of 2nd stage obj. function	2,904.1						

Notes: The demand-side specification includes (non-group specific) brand and year fixed effects. Price is in €10,000, Horsepower is in 100kW, Weight is in 1,000kg, Fuel cost is in €/100km, and Fuel consumption is in L/100km. Distance is the driving distance to the closest car dealer of each brand, in 10km. Willingness to pay, in €/km, is computed as γ_d/α_d $(\times 1,000)$ for each demographic group. Standard errors are computed using the second-step correction formulae in Newey and McFadden (1994).

Table 4: Estimated own-price elasticities

Description	Mean	Std. dev.	10th pct.	Median	90th pct.	Observations
Group 1: Young/Poor	-7.10	3.41	-11.02	-6.35	-3.60	4,960
Group 2: Young/Rich	-6.67	3.17	-10.31	-5.98	-3.43	4,960
Group 3: Middle/Poor	-6.06	2.82	-9.30	-5.44	-3.17	4,960
Group 4: Middle/Rich	-5.36	2.41	-8.14	-4.83	-2.86	4,960
Group 5: Old/Poor	-6.14	2.85	-9.42	-5.51	-3.19	4,960
Group 6: Old/Rich	-4.63	1.99	-6.92	-4.19	-2.55	4,960

Notes: To maintain comparability, statistics use uniform weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups.

Table 3 shows significant heterogeneity in price sensitivities across demographic groups. Price sensitivities vary between −3.440 (young/poor) and −2.021 (old/rich), and the associated median own-price elasticities range from −6.35 to −4.19 (see Table 4). Two intuitive patterns emerge. First, within each age group, high-income consumers are less price sensitive than low-income consumers. Second, younger consumers tend to be the most price sensitive, while middle-aged and older consumers are less so. This age pattern is particularly clear within the high-income category, where price sensitivities decrease monotonically with age (−3.194, −2.443, −2.021). Within the low-income category, the pattern is broadly similar, though middle-aged consumers are slightly less sensitive than older ones (−2.845 versus −2.886).

Combining the estimated distance and price parameters, consumers' willingness to pay to reduce travel distance by one kilometer ranges from €5.3 (young/rich) to €25.9 (old/rich). Young consumers have the lowest willingness to pay, while middle-aged and old/rich consumers have the highest. Importantly, "search costs" should be interpreted broadly to encompass the full buying experience, including pre-purchase visits (e.g., test drives) and expected future visits (e.g., maintenance, after-sale services).[24]

Table 5 summarizes the estimated transaction prices, where all statistics use uniform weights based on total sales to ensure comparability across groups. In line with D'Haultfœuille et al. (2019), we find evidence of third-degree price discrimination. Group 6 (old/rich), the most price-inelastic group, is estimated to always pay the list price. For the other groups, discounts range from 5.0% to 11.9% (corresponding to €978 and €2,347), with the most price-elastic group—group 1 (young/poor)—receiving the largest discounts.

[24] Our estimates are significantly lower than those by Nurski and Verboven (2016), who find a willingness to pay of €112 per kilometer for the Belgian car market in 2011-2012. Although it is difficult to pin down the exact reasons for these differences, there are several contributing factors, from the different models and estimation methods to the different data used for estimation (e.g., the average distance to car dealers in Belgium is substantially smaller than in France, 11.7km versus 16.3km).

Table 5: Estimated transaction prices

Description	Mean	Std. dev.	10th pct.	Median	90th pct.	Observations
Panel A: Transaction price (€)						
Group 1: Young/Poor	20,653	9,914	10,485	18,466	32,032	4,960
Group 2: Young/Rich	20,906	9,910	10,739	18,723	32,274	4,960
Group 3: Middle/Poor	21,352	9,893	11,153	19,140	32,680	4,960
Group 4: Middle/Rich	22,022	9,880	11,794	19,795	33,322	4,960
Group 5: Old/Poor	21,350	9,881	11,104	19,113	32,662	4,960
Group 6: Old/Rich	23,000	9,854	12,768	20,850	34,285	4,960
Panel B: Discount (€)						
Group 1: Young/Poor	2,347	233	2,073	2,392	2,643	4,960
Group 2: Young/Rich	2,094	217	1,841	2,135	2,377	4,960
Group 3: Middle/Poor	1,648	170	1,449	1,677	1,887	4,960
Group 4: Middle/Rich	978	121	844	982	1,151	4,960
Group 5: Old/Poor	1,649	117	1,506	1,658	1,805	4,960
Group 6: Old/Rich	0	0	0	0	0	4,960
Panel C: Discount (%)						
Group 1: Young/Poor	11.92	4.73	6.36	11.43	18.73	4,960
Group 2: Young/Rich	10.64	4.24	5.66	10.18	16.73	4,960
Group 3: Middle/Poor	8.36	3.29	4.45	8.08	13.11	4,960
Group 4: Middle/Rich	4.96	1.98	2.60	4.80	7.78	4,960
Group 5: Old/Poor	8.36	3.23	4.61	7.98	13.00	4,960
Group 6: Old/Rich	0	0	0	0	0	4,960

Notes: All monetary values are in 2018 euros. The demographic group that is estimated to pay the list price is group 6 for all products. Statistics use uniform weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups.

Price discrimination versus search costs. Before introducing the online distribution channel, we perform counterfactuals to shed light on the relationship between price discrimination and search costs (Table 6 and Table 7). We consider three scenarios: (i) no price discrimination, (ii) reduced search costs (by 37%, matching our Tesla estimates, and by 100%), and (iii) both combined.

Under price discrimination, young and low-income consumers benefit from discounts, with gains in consumer surplus ranging from €74.5 to €113.3 per consumer per year. In contrast, groups 4 (middle/rich) and 6 (old/rich) face higher prices under price discrimination and lose €30.1 and €321 per consumer per year, respectively. On aggregate, given the large share of purchases from older and wealthier groups, price discrimination decreases consumer surplus by €14.7 per consumer per year. Industry profits increase by 3.3%. These results suggest that price discrimination is profitable for firms and redistributive for consumers.

Table 6: Effect of price discrimination and search costs on purchases

	Baseline	No discr.	Red. search cost		No discr. + Red. search cost	
			-37%	-100%	-37%	-100%
Panel A: Transaction prices, uniform weights						
Group 1: Young/Poor	21,418		21,419	21,420		
Group 2: Young/Rich	21,665		21,666	21,666		
Group 3: Middle/Poor	22,097		22,100	22,106		
Group 4: Middle/Rich	22,750		22,753	22,759		
Group 5: Old/Poor	22,099		22,104	22,112		
Group 6: Old/Rich	23,733		23,738	23,747		
Uniform price		22,650			22,652	22,656
Panel B: Transaction prices, sales-weighted						
Group 1: Young/Poor	20,350		20,401	20,523		
Group 2: Young/Rich	22,113		22,134	22,186		
Group 3: Middle/Poor	21,929		22,045	22,301		
Group 4: Middle/Rich	24,106		24,182	24,355		
Group 5: Old/Poor	20,600		20,685	20,852		
Group 6: Old/Rich	23,644		23,724	23,891		
Uniform price		22,435			22,504	22,657
Panel C: Sales, in units						
Group 1: Young/Poor	109,103	-32,987	+2,160	+6,113	-31,446	-28,615
Group 2: Young/Rich	78,590	-17,460	+682	+1,893	-16,933	-15,996
Group 3: Middle/Poor	143,096	-16,623	+4,888	+14,020	-12,123	-3,669
Group 4: Middle/Rich	238,844	+5,335	+4,965	+13,976	+10,438	+19,703
Group 5: Old/Poor	112,339	-14,245	+3,411	+9,590	-11,084	-5,321
Group 6: Old/Rich	309,327	+47,684	+5,975	+16,734	+54,273	+66,091
All consumers	991,299	-28,296	+22,081	+62,326	-6,875	+32,193
Panel D: Average distance to car models, in km						
Group 1: Young/Poor	17.00	+0.23	+0.40	+1.22	+0.63	+1.48
Group 2: Young/Rich	16.46	+0.19	+0.14	+0.44	+0.33	+0.64
Group 3: Middle/Poor	18.85	+0.12	+0.74	+2.24	+0.86	+2.39
Group 4: Middle/Rich	15.45	+0.04	+0.38	+1.22	+0.42	+1.26
Group 5: Old/Poor	20.70	+0.02	+0.58	+1.66	+0.61	+1.71
Group 6: Old/Rich	14.65	-0.05	+0.42	+1.28	+0.36	+1.20

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome (column 1). All monetary values are in 2018 euros. Uniform-weighted transaction prices (Panel A) use weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups and counterfactual experiments. Sales-weighted transaction prices (Panel B) use realized sales for each demographic group and counterfactual experiment. For sales and average distances, we report values at baseline in the first column and changes from baseline in the other columns.

Table 7: Effect of price discrimination and search costs on welfare

	No discr.	Red. search cost		No discr. + Red. search cost	
		-37%	-100%	-37%	-100%
Panel A: Total change in consumer surplus, in € per consumer					
Group 1: Young/Poor	-74.5	+5.0	+14.2	-71.0	-64.6
Group 2: Young/Rich	-89.7	+3.7	+10.1	-87.0	-82.2
Group 3: Middle/Poor	-81.5	+24.7	+71.2	-59.4	-17.5
Group 4: Middle/Rich	+30.1	+28.5	+80.4	+59.6	+113.4
Group 5: Old/Poor	-113.3	+27.8	+79.0	-88.3	-42.3
Group 6: Old/Rich	+321.0	+39.5	+111.3	+367.6	+452.1
All consumers	+14.7	+21.1	+59.6	+36.1	+75.4
Panel B: Change in profits					
Profits, in million €	-135.4	+95.0	+267.1	-44.3	+120.5
Profits, in %	-3.3	+2.3	+6.4	-1.1	+2.9

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome. All monetary values are in 2018 euros. Profits are in million euros.

Reducing search costs has little effect on pricing decisions (Panel A of Table 6): firms capture the gains through market expansion rather than higher prices. On the demand side, lower search costs lead consumers to switch to more expensive car models (Panel B) and to brands whose dealers are located farther away (Panel D). Eliminating search costs entirely increases consumer surplus by €59.6 per consumer-year and industry profit by 6.4%, making search cost elimination roughly 4 times as valuable as price discrimination for consumers and twice as valuable for firms.

Finally, removing price discrimination and reducing search costs simultaneously benefits consumers but has an ambiguous effect on firms (columns 4–5 of Table 7). A 37% reduction in search costs is not large enough for market expansion to overcome the losses from uniform pricing, and firm profits fall by 1.1%. Eliminating search costs entirely, however, generates a 2.9% increase in aggregate profits. These results suggest that the profitability of an online distribution channel hinges on its ability to achieve substantial search cost reductions.

6 Model with the online distribution channel

In our counterfactual analysis, we extend the model to allow consumers to choose between purchasing in person or online. Online purchases are made directly from the producer's website, as currently observed for Tesla. In-person purchases occur at the closest dealer of the chosen brand, as in our baseline model. Purchasing online reduces search costs, as consumers have the car delivered to their home without visiting a dealer. They do, however, pay the list price posted on the website. Purchasing in person requires searching for dealers and visiting them, but allows consumers to obtain a price below the list price. As in the baseline model, Tesla charges the list price in both channels.

Throughout, we remain agnostic about the extent to which buying online reduces consumers' search costs. If producers also handled delivery and pick-up for test drives and maintenance, purchasing online could eliminate search costs entirely; in more realistic cases, some search costs may remain if consumers still visit car dealers, e.g., for test drives and maintenance. We investigate these extremes and intermediate scenarios by repeating the analysis for different levels of reduction in average search costs, captured by the parameter $\rho \in [0, 1]$. Note that, while ρ captures a potentially different reduction in average search costs than the τ_d's we estimate for Tesla (other producers may implement an online distribution channel with different features than Tesla's), we will present the case of $\rho = \sum_d \phi_d \widehat{\tau}_d$ as a baseline, as if all other producers were to mimic Tesla's online distribution channel.

We focus on the case in which producers do not price discriminate against consumers who purchase online. We do this to mimic the stated intentions of Ford and the observed practice of Tesla, which lean heavily toward price transparency and streamlining the transaction process (see the Introduction). However, for completeness, we also perform a set of counterfactuals with price discrimination in both distribution channels (always maintaining that Tesla does not price discriminate on any distribution channel). The results of these additional counterfactuals are reported in Appendix Appendix D.

Lastly, we assume that the introduction of an online distribution channel does not change the observed configuration of car dealers (i.e., no entry or exit of car dealers) and that it does not lead to any savings on marginal costs (which could instead materialize when bypassing the "middleman") or additional logistical costs (e.g., consumer-specific car delivery or pick up). These assumptions imply that our results should be interpreted as short-run responses to the introduction of an online distribution channel. Although we do not explicitly model these exit decisions and cost reductions, we conduct a series of additional counterfactuals in Section 7.5 to provide some insight into these potentially important long-run responses of the industry.

6.1 Demand

We extend our model from Sections 2.1-2.2 to allow consumers to choose, for each producer, whether to search in person or online. This captures the idea that consumers trade off potentially lower in-person prices (due to discounts) against lower online search costs, and that different producers may be searched via different channels by the same consumer.

Recall that we have two types of consumers: those willing to purchase cars online (probability ψ_d) and those who are not $(1-\psi_d)$. People not willing to purchase cars online search and purchase cars in person, following the Moraga-González et al. (2023) search protocol summarized in Sections 2.1-2.2. In contrast, people willing to purchase cars online can, for each producer f, choose whether to search and purchase in person or online. We denote the in-person channel by P and the online channel by O. For clarity, we define p^P_{jd} as the discriminatory in-person price paid for car model j by consumers of group d, and $p^O_j = \overline{p}_j$ as the list price paid online for j by all consumers. All car models are available in both distribution channels.

Channel choice. Before making any decisions, consumer i of group d living in municipality m knows: (a) the location of the closest car dealer of each producer f and the subset of car models each sells, (b) all prices p^P_{jd} and p^O_j for all j, and hence all elements of indirect utility (1)—evaluated at channel-specific prices—but not ϵ_{ijdm} for each j, (c) ϵ_{i0dm} and the distribution of all ϵ_{ijdm}'s, and (d) the distribution of search costs for each producer f in each channel $\ell \in \{P, O\}$, which is the Gumbel-preserving distribution (5) with channel-specific mean $\overline{\mathrm{sc}}^{f,\ell}_{dm}$. In contrast, at the time of choosing channels, consumer i does not know the exact realization of search costs sc^ℓ_{ifdm}.

Given this information structure, for each producer f, consumer i chooses the channel ℓ that maximizes the expected value of the channel-specific summary statistic

$$\mathrm{w}^\ell_{ifdm} \equiv \min\left\{ r^\ell_{ifdm},\ \max_{j\in\mathcal{G}_f}\{U^\ell_{ijdm}\}\right\}, \tag{19}$$

where U^ℓ_{ijdm} equals (1) evaluated at the channel-specific price p^P_{jd} for $\ell = P$ or p^O_j for $\ell = O$, and the reservation value r^ℓ_{ifdm} solves $H_0(r - \delta^\ell_{ifdm}) = \mathrm{sc}^\ell_{ifdm}$ as in (2). To compute the expectation of w^ℓ_{ifdm}, note that under (5) applied to sc^ℓ_{ifdm} with mean $\overline{\mathrm{sc}}^{f,\ell}_{dm}$, w^ℓ_{ifdm} is distributed Gumbel$(\delta^\ell_{ifdm} - \overline{\mathrm{sc}}^{f,\ell}_{dm}, 1)$ with

$$\delta^\ell_{ifdm} \equiv \ln\left(\sum_{h\in\mathcal{G}_f} \exp(\delta^\ell_{hd} + \mu_{hdm}(\nu_i))\right), \tag{20}$$

the channel-specific inclusive value for producer f and δ^ℓ_{hd} evaluated at p^P_{hd} for $\ell = P$ or at p^O_h for $\ell = O$. Since the expected value of w^ℓ_{ifdm} equals the location of its Gumbel distribution plus the Euler constant, the chosen channel for producer f is:

$$\ell^*_{ifdm} \in \arg \max_{\ell \in \{P,O\}} \left\{ \delta^{\ell}_{ifdm} - \overline{\mathrm{sc}}^{f,\ell}_{dm} \right\}. \tag{21}$$

Equivalently, (21) selects the channel whose w^{ℓ}_{ifdm} distribution first-order stochastically dominates the other (since both have scale 1).

The channel-specific mean search costs for $f \neq$ tesla are $\overline{\mathrm{sc}}^{f,P}_{dm} = -\gamma_d \,\mathrm{dist}_{f,m}$ for in-person search and $\overline{\mathrm{sc}}^{f,O}_{dm} = -\rho\, \gamma_d \,\mathrm{dist}_{f,m}$ for online search, with $\rho \in [0,1]$ capturing the reduction in average search costs from shopping online. Rearranging (21), consumer i searches producer $f \neq$ tesla online if and only if:

$$\underbrace{-(1-\rho)\,\gamma_d \,\mathrm{dist}_{f,m}}_{\text{online search cost savings}} \;\geq\; \underbrace{\delta^{P}_{ifdm} - \delta^{O}_{ifdm}}_{\text{in-person price savings}}. \tag{22}$$

Since $\gamma_d \leq 0$, the left-hand side of (22) is non-negative. The right-hand side is positive when discounts are offered in person. Intuitively, producers that are far from the consumer's municipality are more likely to be searched online, while producers offering large in-person discounts are more likely to be searched in person. The channel choice is binding: once consumer i commits to searching producer f via channel ℓ^*_{ifdm}, the transaction price for any $j \in \mathcal{G}_f$ is determined by that channel.

For Tesla, we maintain the same treatment as in the baseline model of Section 2.2. Tesla does not price discriminate: $p^P_{jd} = p^O_j = \overline{p}_j$ for all $j \in \mathcal{G}_{\text{tesla}}$ and all d. The average search costs for Tesla are $\overline{\mathrm{sc}}^{\text{tesla},P}_{dm} = -\gamma_d \,\mathrm{dist}_{\text{tesla},m}$ for in-person and $\overline{\mathrm{sc}}^{\text{tesla},O}_{dm} = -\tau_d\, \gamma_d \,\mathrm{dist}_{\text{tesla},m}$ for online, with $\tau_d \in [0,1]$ as estimated in the baseline model. Due to uniform prices across distribution channels and lower online average search costs, consumers willing to purchase cars online always choose the online channel: condition (22) for Tesla simplifies to $-(1-\tau_d)\,\gamma_d \,\mathrm{dist}_{\text{tesla},m} \geq 0$, which is always satisfied given that $\gamma_d \leq 0$.

Search and purchase. Given the channel choices $(\ell^*_{ifdm})_{f \in \mathcal{F}}$, consumer i observes the realized search costs $\mathrm{sc}^{\ell^*_{ifdm}}_{ifdm}$ for $f \in \mathcal{F}$ as drawn from the corresponding Gumbel-preserving distributions (5). The consumer then follows the Moraga-González et al. (2023) search protocol: she computes reservation values $r^{\ell^*_{ifdm}}_{ifdm}$ for each producer using (2), searches producers sequentially in order of reservation values, learns ϵ_{ijdm} for all $j \in \mathcal{G}_f$ upon searching f through channel ℓ^*_{ifdm}, and stops when the best option found exceeds the next reservation value. The outcome of this search is "as if" consumer i observed $\mathrm{w}^{\ell^*_{ifdm}}_{ifdm}$ for all f and selected the producer corresponding to the maximum.

The idiosyncratic taste shock ϵ_{ijdm} for car model j is the same regardless of the channel through which j's producer is searched. This reflects the fact that ϵ_{ijdm} captures the consumer's match value with the car itself—its feel, handling, and fit with the consumer's tastes—which does not depend on the distribution channel.

Purchase probabilities. Given the channel choices $(\ell^*_{ifdm})_{f\in\mathcal{F}}$ and the Gumbel assumptions on ϵ_{ijdm} and $\mathrm{w}_{ifdm}^{\ell^*_{ifdm}}$, the purchase probability of consumer i of demographic group d living in municipality m who is willing to purchase cars online is:

$$\Pr_{dm}(j \mid \nu_i) = \frac{\exp\left(V_{jdm}(\nu_i)\right)}{1 + \sum_{k=1}^{J} \exp\left(V_{kdm}(\nu_i)\right)}, \tag{23}$$

where, for $j \in \mathcal{G}_f$ with $f \neq$ tesla,

$$V_{jdm}(\nu_i) = \delta_{jd}^{\ell^*_{ifdm}} + \mu_{jdm}(\nu_i) - \overline{\mathrm{sc}}_{dm}^{f,\ell^*_{ifdm}}, \tag{24}$$

and ℓ^*_{ifdm} is the chosen channel for producer f as defined in (21). For $j \in \mathcal{G}_{\text{tesla}}$, as in the baseline model of Section 2.2, the online channel is always chosen and

$$V_{jdm}(\nu_i) = \delta_{jd} + \mu_{jdm}(\nu_i) + \tau_d\, \gamma_d\, \mathrm{dist}_{\text{tesla},m}, \tag{25}$$

where we denote δ_{jd} without channel superscript since Tesla does not price discriminate. As in the baseline model, all choices are deterministic from the consumer's perspective and purchase probabilities arise only from the researcher's perspective because $\mathrm{w}_{ifdm}^{\ell^*_{ifdm}}$ and ϵ_{ijdm} are unobserved.

For consumers unwilling to purchase cars online, the purchase probability is as in the baseline model (6), which we denote by $\Pr^P_{dm}(j \mid \nu_i)$ to distinguish it from (23). These consumers face average search costs $-\gamma_d\, \mathrm{dist}_{fm}$ for all producers, including Tesla, and in-person prices p^P_{jd} for all j (with $p^P_{jd} = \overline{p}_j$ for $j \in \mathcal{G}_{\text{tesla}}$).

From (23), the probability that a consumer willing to purchase online buys car model j from channel ℓ is $\Pr_{dm}(j, \ell \mid \nu_i) = \Pr_{dm}(j \mid \nu_i) \cdot \mathbb{1}\left\{\ell^*_{if(j)dm} = \ell\right\}$, where $\mathbb{1}\{\cdot\}$ is the indicator function and $\ell^*_{if(j)dm}$ depends on ν_i through δ^{ℓ}_{ifdm}. Integrating over ν_i and accounting for the share ψ_d willing to purchase online, the channel-specific purchase probability for group d in municipality m is:

$$s^P_{jdm}(p^P_d, p^O) = \psi_d \cdot \int \mathrm{Pr}_{dm}(j \mid \nu_i) \cdot \mathbb{1}\left\{\ell^*_{if(j)dm} = P\right\} dF(\nu_i)$$
$$+ (1-\psi_d) \cdot \int \mathrm{Pr}^P_{dm}(j \mid \nu_i)\, dF(\nu_i), \tag{26}$$

$$s^O_{jdm}(p^P_d, p^O) = \psi_d \cdot \int \mathrm{Pr}_{dm}(j \mid \nu_i) \cdot \mathbb{1}\left\{\ell^*_{if(j)dm} = O\right\} dF(\nu_i), \tag{27}$$

where $p^P_d = (p^P_{1d}, ..., p^P_{Jd})$ and $p^O = (p^O_1, ..., p^O_J)$. The total municipal-level purchase probability is $s_{jdm}(p^P_d, p^O) = s^P_{jdm}(p^P_d, p^O) + s^O_{jdm}(p^P_d, p^O)$. Averaging over municipalities, we obtain the national-level market share of group d for car model j from distribution channel $\ell \in \{P, O\}$:

$$s^\ell_{jd}(p^P_d, p^O) = \sum_{m \in \mathcal{M}} w_{dm} \cdot s^\ell_{jdm}(p^P_d, p^O), \tag{28}$$

where $w_{dm} \equiv M_{dm}/M_d$ is defined as in (8).

Consistency with the baseline model. The counterfactual model described above is consistent with the baseline model of Sections 2.1-2.2. When only the in-person channel is available, the channel choice step (21) is degenerate: $\ell^*_{ifdm} = P$ for all f. In this case, the channel choice stage is vacuous, and consumers proceed as in the baseline protocol. Consequently, (23) reduces to (6) and the counterfactual model simplifies to the baseline model.

6.2 Supply

As in Section 2.3, we consider a Bertrand-Nash price-setting game in which every producer f chooses transaction prices for in-person sales $p^P_j = (p^P_{j1}, ..., p^P_{jD})$ and the non-discriminatory online price p^O_j (i.e., the list price) for each j they sell by maximizing the national-level profit function

$$\pi_f(p^P, p^O) = \sum_{d=1}^{D} \phi_d \sum_{j \in \mathcal{G}_f} s^P_{jd}(p^P_d, p^O) \cdot (p^P_{jd} - c_j) + \sum_{d=1}^{D} \phi_d \sum_{j \in \mathcal{G}_f} s^O_{jd}(p^P_d, p^O) \cdot (p^O_j - c_j), \tag{29}$$

where $p^P = (p^P_1, ..., p^P_D)$ collects all in-person prices and $p^O = (p^O_1, ..., p^O_J)$ collects all online prices, and the national-level group-specific market shares $s^P_{jd}(p^P_d, p^O)$ and $s^O_{jd}(p^P_d, p^O)$ correspond to (28) for $\ell \in \{P, O\}$. For Tesla, (29) involves fewer prices to be chosen, since $p^P_{jd} = p^O_j$ for all $j \in \mathcal{G}_{\text{tesla}}$ and $d = 1, ..., D$.

7 Counterfactual simulations

In this section, we present the results of our main counterfactual experiments. We simulate scenarios in which an online distribution channel is introduced in the French car industry. All counterfactuals are performed on our 2019 data. Unless otherwise stated, consumers purchasing in person at car dealers can receive discounts as a result of price discrimination and incur full search costs. Consumers who purchase online instead pay the list prices posted on the car manufacturers' websites and face a fraction ρ of the in-person search costs, with $\rho = \sum_d \phi_d \widehat{\tau}_d$.

We treat Tesla differently from other manufacturers, assuming that Tesla charges list prices in both channels, $p^P_{jd} = p^O_j = \overline{p}_j$ for all $j \in \mathcal{G}_{tesla}$, and continues to offer the same level of online search cost reductions as estimated above.

We present the results of our counterfactuals in Table 8. Column (1) reports the baseline without online sales (with the exception of Tesla), and columns (2)–(5) report the results from our four counterfactual experiments. Our counterfactuals vary along two dimensions. First, we consider two levels of search cost reductions: a 37% reduction, corresponding to the average reduction estimated from Tesla, and a 100% reduction which would occur if firms also handled deliveries and pick-ups for test drives and maintenance. This allows us to assess the role that search cost savings play in the rollout of the online channel. Second, we consider two levels of "access" to the online distribution channel. In our preferred specification, for each demographic group d, access is limited to a share ψ_d of consumers who are willing to purchase cars online (according to our survey of attitudes, see Table 1). Our alternative specification considers universal access to the online channel (i.e., $\psi_d = 1$ for each d). This allows us to quantify the impact of expanding access to the online channel, which may occur over time as consumers become more accustomed to purchasing cars online.

7.1 Limited access to the online distribution channel

The results of our counterfactuals with limited access to the online channel are presented in columns (2) and (3) of Table 8. These results highlight three main patterns.

The first pattern concerns firms' pricing strategies. Two effects shape the pricing equilibrium with online sales. The first is a *competitive effect*: by reducing search costs, the online channel allows firms whose car dealers were previously located far from consumers to compete more aggressively. This intensifies price competition and drives list prices down, mostly benefiting inelastic consumers who receive small or no discounts in person.

Table 8: Introducing the online distribution channel

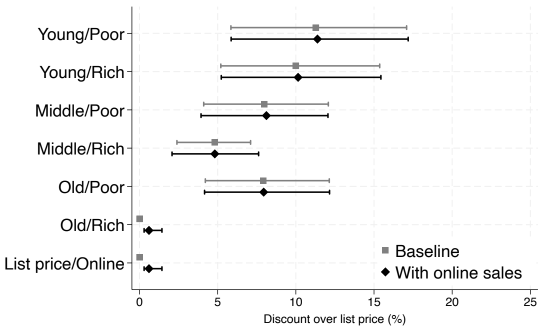
	No online channel	Limited online access		Universal online access	
		-37%	-100%	-37%	-100%
Panel A: Transaction prices, uniform weights					
Group 1: Young/Poor	21,418	21,405	21,427	21,388	21,485
Group 2: Young/Rich	21,665	21,642	21,645	21,606	21,647
Group 3: Middle/Poor	22,097	22,090	22,300	22,487	22,751
Group 4: Middle/Rich	22,750	22,750	22,916	22,854	22,751
Group 5: Old/Poor	22,099	22,084	22,205	22,487	22,751
Group 6: Old/Rich	23,733	23,562	23,217	22,991	22,751
Online/List price		23,562	23,217	22,991	22,751
Panel B: Transaction prices, sales-weighted					
Group 1: Young/Poor	20,350	20,295	20,098	20,137	19,543
Group 2: Young/Rich	22,113	22,057	21,913	21,932	21,392
Group 3: Middle/Poor	21,929	21,655	21,357	19,814	21,420
Group 4: Middle/Rich	24,106	23,592	23,745	23,466	23,271
Group 5: Old/Poor	20,600	20,511	20,403	18,963	20,315
Group 6: Old/Rich	23,644	23,554	23,205	22,613	21,821
Online/List price		31,297	29,570	27,268	25,673
Panel C: Sales, in units					
Group 1: Young/Poor	109,103	+381	-401	+729	-2,099
Group 2: Young/Rich	78,590	+423	+185	+812	-444
Group 3: Middle/Poor	143,096	-128	-5,926	-9,964	-9,059
Group 4: Middle/Rich	238,844	-1,643	-4,052	-1,273	+11,651
Group 5: Old/Poor	112,339	+441	-2,065	-6,031	-8,535
Group 6: Old/Rich	309,327	+7,801	+25,676	+36,783	+58,547
All consumers	991,299	+7,275	+13,417	+21,056	+50,061
Panel D: Proportion of online sales					
Group 1: Young/Poor	0	0.001	0.010	0.009	0.073
Group 2: Young/Rich	0	0.002	0.010	0.018	0.072
Group 3: Middle/Poor	0	0.014	0.116	0.384	0.735
Group 4: Middle/Rich	0	0.076	0.223	0.516	0.684
Group 5: Old/Poor	0	0.004	0.036	0.312	0.733
Group 6: Old/Rich	0	0.179	0.216	0.576	0.679
Panel E: Average distance to car models, in km					
Group 1: Young/Poor	17.00	-0.08	-0.15	-0.13	-0.07
Group 2: Young/Rich	16.46	-0.04	-0.10	-0.08	-0.15
Group 3: Middle/Poor	18.85	-0.16	+0.27	+0.02	+2.45
Group 4: Middle/Rich	15.45	-0.03	+0.53	+0.39	+1.30
Group 5: Old/Poor	20.70	-0.06	-0.06	-0.28	+1.69
Group 6: Old/Rich	14.65	+0.24	+0.59	+0.51	+1.22

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome (column 1). In-person sales assume price discrimination and full search costs, while online sales assume list prices and reduced search costs. Access to the online channel in columns (2)–(3) is limited to consumers willing to purchase cars online according to our survey instrument (see Table 1); access is universal in columns (4)–(5). All monetary values are in 2018 euros. Uniform-weighted transaction prices (Panel A) use weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups and counterfactual experiments. Sales-weighted transaction prices (Panel B) use realized sales for each demographic group and counterfactual experiment. For sales and average distances, we report values at baseline in the first column and changes from baseline in the other columns.

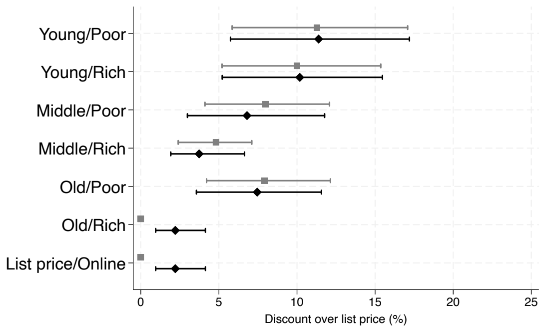
The second is a *online steering effect*: as search costs decrease, firms have an incentive to raise in-person prices to nudge consumers toward the online channel. Since online transactions take place at the list price (without discounts), they are more profitable per unit than in-person sales for groups that would otherwise receive discounts.

To better visualize these price changes, Figure 6 plots the distribution of list prices and in-person discounts for the limited access cases. Discounts are measured at the product level and normalized by each product's list price in the no-online-sales baseline. For each demographic group, we report the median discount and the 10th-to-90th percentile range across all products. In both panels, the most visible change relative to baseline concerns the list price: the introduction of the online channel lowers list prices, with a larger decrease in panel (b) when search costs are fully eliminated. This reflects the competitive effect discussed above. For most demographic groups, the in-person discount distributions remain close to baseline, confirming that the online channel has limited direct impact on in-person pricing for groups who do not access it widely. The exception is panel (b), where the discounts of groups 3, 4, and 5 are slightly compressed as firms raise in-person prices in response to the online steering effect.

Figure 6: Price dispersion with limited access to the online channel



(a) -37% search costs



(b) No search costs

Notes: These figures illustrate price dispersion in the in-person channel given limited access to the online channel, for varying levels of search cost reductions as per Table 8. Price dispersion is represented as a discount over the list price at baseline (as observed in the data), in percentage points. The marker represents the median discount, and the brackets represent the 10th–90th percentile range. The baseline prices are shown in gray and the counterfactual experiments with online sales are shown in black.

The second pattern concerns sales. With limited access, the online channel introduces a modest market expansion of 0.7–1.4% (depending on the reduction in search costs), almost entirely driven by old/rich consumers (group 6). The online channel provides these consumers with both lower prices—list prices decrease, and these consumers do not receive discounts in person—and search cost savings, generating strong substitution away from the outside option. For all other groups, the online list price exceeds the discounted in-person price, so the price and search cost effects go in opposite directions.

The impact on sales is nevertheless heterogeneous across groups. Because of their relatively low search costs (small estimates for γ_d in absolute value) and relatively high price sensitivity (large estimates for α_d), young consumers (groups 1 and 2) are largely unaffected. The search cost savings of the online channel do not compensate for the price premium of switching online (as they would lose the in-person discounts). Groups 3 (middle-aged/poor) and 5 (old/poor) experience small effects when the search cost reduction is modest—for reasons similar to those of young consumers—but sales decline substantially when search costs are fully eliminated. When this occurs, the online channel becomes attractive to a share of these consumers, prompting firms to raise in-person prices. As a result, these consumers face higher average prices, reducing total purchases. Middle-aged/rich consumers (group 4) decrease their purchases in both scenarios. Since their in-person prices are close to the list prices, the availability of the online channel leads some consumers to search online, even with moderate reductions in search costs. From the online channel, they opt for the outside option more frequently, reducing total purchases. This effect compounds when search costs are fully eliminated, as in-person prices also rise.

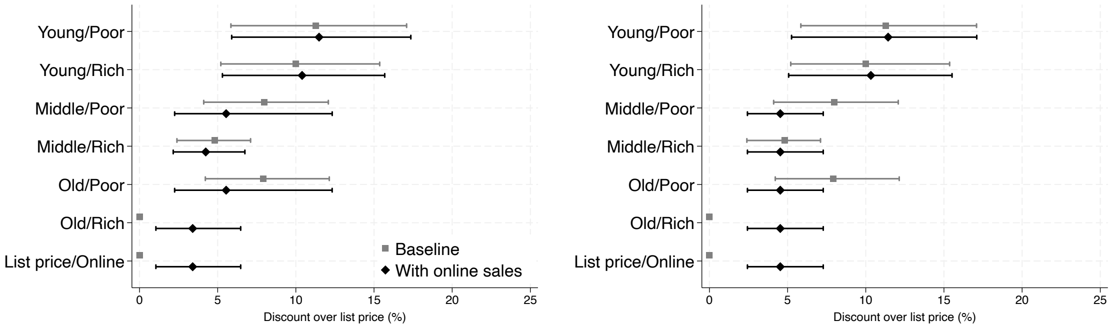
The last pattern concerns the market penetration of the online channel. With limited access, online sales remain modest: around 8% of purchases take place online with a 37% reduction in search costs, rising to around 15% when search costs are fully eliminated. Adoption is highly uneven across groups: groups 1, 2, and 5 (young/poor, young/rich, and old/poor) almost never purchase online under either scenario, as their large in-person discounts outweigh the search cost savings. Because of their relatively low price sensitivity, groups 4 and 6 (middle-aged/rich and old/rich) show the highest adoption, each reaching around 22% of their purchases when search costs are fully eliminated. Finally, consumers who shop online tend to purchase car models whose closest dealer is located farther away. Effectively, the online channel enlarges these consumers' choice sets and enables them to find better matches to their preferences.

7.2 Universal access to the online distribution channel

We now consider universal access to the online channel, where all consumers are willing to purchase online. The results are presented in columns (4)–(5) of Table 8.

Expanding access to the online channel substantially amplifies the three patterns dis-

Figure 7: Price dispersion with universal access to the online channel



(a) -37% search costs

(b) No search costs

Notes: These figures illustrate price dispersion in the in-person channel given universal access to the online channel, for varying levels of search cost reductions as per Table 8. Price dispersion is represented as a discount over the list price at baseline (as observed in the data), in percentage points. The marker represents the median discount, and the brackets represent the 10th–90th percentile range. The baseline prices are shown in gray and the counterfactual experiments with online sales are shown in black.

cussed in Section 7.1. Figure 7 illustrates the larger consequences on equilibrium prices. The *competitive effect* is markedly amplified relative to limited access: with a broader consumer base able to purchase online, competition intensifies and list prices fall more sharply, benefiting all groups and in particular group 6 (old/rich). At the same time, in-person prices rise substantially for groups 3, 4, and 5 (middle/poor, middle/rich, and old/poor), as firms respond to universal access by raising in-person prices so as to redirect these consumers toward the more profitable online channel. The most striking result occurs when search costs are fully eliminated: four out of six groups—groups 3, 4, 5, and 6—end up paying the list price whether they purchase in person or online, so that transaction prices fully converge across channels. Groups 1 and 2 (young/poor and young/rich) are the exceptions, with their in-person prices remaining essentially unchanged as their large baseline discounts continue to outweigh the benefits of switching online. These price changes are the direct manifestation of the *online steering effect*: firms nudge consumers toward the online channel by aligning in-person and list prices, but this equalization is only fully achieved when universal access is combined with a complete elimination of search costs.

Market expansion is considerably larger than under limited access. Total purchases increase by around 2% with a 37% reduction in search costs, and by up to 5% when search costs are fully eliminated (Panel C), compared to 0.7–1.4% under limited access. The gains are still concentrated among old/rich consumers (group 6), whose sales increase is now much larger. In contrast, groups 3 and 5 (middle/poor and old/poor) experience meaningful sales declines, as firms raise their in-person prices to redirect them toward the more profitable online channel. Online adoption rises dramatically: around 40% of all purchases take place online under a 37% search cost reduction, rising to around 59% when search costs are fully eliminated (Panel D), compared to around 8% and 15% under limited access. These figures suggest that expanding consumer access to the online channel—by convincing consumers to consider purchasing online—has a substantially larger effect on online penetration than further reducing search costs alone.

7.3 Only Nissan-Renault introduces an online channel

In the previous sections, we simulated counterfactuals involving the introduction of an online distribution channel by all firms simultaneously. Here, we instead consider a counterfactual scenario in which one large car manufacturer starts selling online while its competitors continue to sell only in person. We choose the Nissan-Renault group as our candidate online firm and re-evaluate counterfactual prices for all firms, for varying levels of search cost reductions.[25]

The results are presented in Table 9. Panel A shows that Nissan-Renault's pricing broadly follows the patterns described in Section 7.1, where all firms introduce an online channel simultaneously. The more striking result is in Panel B: the introduction of an online channel by a single firm exerts competitive pressure on all other manufacturers, whose prices fall across every demographic group and counterfactual. This cross-firm competitive effect is most pronounced under universal access, where the online channel is available to the full consumer base, enabling more consumers to switch between firms and compelling rivals to lower their prices more substantially. Under limited access, fewer consumers can switch between firms through the online channel, and the competitive effect on rivals is correspondingly smaller.

These results also clarify the distinct roles of search costs and list prices in generating competition across firms. Within Nissan-Renault, lower search costs make the online and in-person channels closer substitutes. By lowering list prices, Nissan-Renault can profitably induce consumers to switch to the online channel. However, across firms, this within-firm channel reallocation does not readily propagate: consumers switching between Nissan-Renault's in-person and online channels does not greatly affect rivals' prices (and market shares, not reported in Table 9). Taken together, these results, com-

[25]The Nissan-Renault group also includes Dacia and Mitsubishi and had a share of around 27% of the French market during our sample period.

Table 9: Only Nissan-Renault introduces online channel

	No online channel	Limited online access		Universal online access	
		-37%	-100%	-37%	-100%
Panel A: Transaction prices, Nissan-Renault group					
Group 1: Young/Poor	16,673	16,658	16,666	16,651	16,663
Group 2: Young/Rich	16,922	16,895	16,898	16,858	16,864
Group 3: Middle/Poor	17,377	17,348	17,393	17,872	18,052
Group 4: Middle/Rich	18,035	17,981	18,073	18,309	18,052
Group 5: Old/Poor	17,397	17,375	17,404	17,338	18,052
Group 6: Old/Rich	19,044	18,946	18,789	18,309	18,052
Online/List price		18,946	18,789	18,309	18,052
Panel B: Transaction prices, other manufacturers					
Group 1: Young/Poor	23,248	23,236	23,236	23,227	23,227
Group 2: Young/Rich	23,494	23,475	23,475	23,460	23,460
Group 3: Middle/Poor	23,917	23,895	23,895	23,863	23,863
Group 4: Middle/Rich	24,568	24,529	24,529	24,474	24,474
Group 5: Old/Poor	23,913	23,897	23,897	23,835	23,835
Group 6: Old/Rich	25,541	25,488	25,488	25,379	25,378

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome (column 1). This table presents the equilibrium transaction prices when only the Nissan-Renault group introduces an online distribution channel; competitors are restricted to in-person transactions (as observed in the data). All monetary values are in 2018 euros. Transaction prices use uniform weights $w_j = \sum_d \phi_d s_{jd} / \sum_{k \in \mathcal{J}_g} \sum_d \phi_d s_{kd}$, where $\mathcal{J}_g$ is the set of car models offered by group $g \in \{\text{Nissan-Renault}, \text{Other}\}$, constructed from total sales in the baseline scenario and fixed across demographic groups and counterfactual experiments.

bined with those from Section 7.1, suggest that the introduction of an online channel has two effects on competition: a within-firm effect, through which lower search costs reallocate consumers across channels, and a between-firm effect, through which the expansion of the online market to a larger consumer base intensifies competition for market share and compels rivals to lower their prices.

7.4 Welfare analysis

We now turn to the welfare consequences of introducing an online distribution channel. Table 10 reports the aggregate effects on consumer surplus and profits for all counterfactuals, varying both the level of online access and the reduction in search costs.

Consumer surplus. Panel A of Table 10 shows that average consumer surplus increases in all scenarios, ranging from €12.1 per consumer under limited access with a 37% search cost reduction to €87.4 under universal access with no search costs. These aggregate gains, however, mask substantial heterogeneity. The online channel is redis-

Table 10: Effect of online channel on welfare

	Limited online access		Universal online access	
	-37%	-100%	-37%	-100%
Panel A: Change in consumer surplus, in € per consumer				
Group 1: Young/Poor	+1.0	-0.4	+2.5	-3.0
Group 2: Young/Rich	+2.6	+2.2	+6.9	+1.6
Group 3: Middle/Poor	+1.4	-25.5	-41.9	-34.8
Group 4: Middle/Rich	-0.7	-13.8	+5.9	+85.9
Group 5: Old/Poor	+4.6	-14.3	-40.2	-56.0
Group 6: Old/Rich	+56.1	+179.3	+262.8	+427.1
All consumers	+12.1	+27.5	+43.6	+87.4
Panel B: Change in profits				
Profits, in million €	-1.5	+24.4	+3.1	+136.8
Profits, in %	-0.04	+1.5	+0.1	+3.3

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome, without online sales. All monetary values are in 2018 euros. Profits are in million euros.

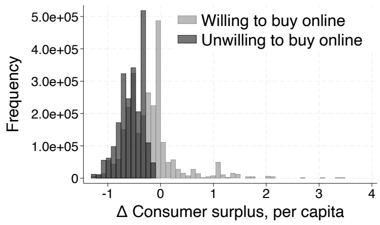
tributive: older and wealthier consumers gain the most, benefiting from both lower list prices (competitive effect) and significant search cost savings, while younger and poorer consumers can experience losses when in-person price increases driven by the online steering effect outweigh their search cost savings.

Figure 8 plots the distribution of the change in consumer surplus by demographic group, distinguishing between consumers who are willing to purchase cars online and those who are not. We focus on the scenario with full elimination of search costs, which represents the most favorable scenario for the online channel.

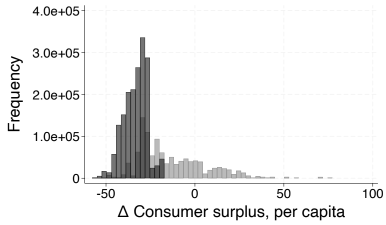
The figures reveal stark differences between and within demographic groups. Consumers willing to purchase online gain from both lower search costs and the competitive effect on prices, while those unwilling to do so are only affected through prices. As a result, within each group, consumers willing to purchase online tend to fare better than those who are not. Across groups, heterogeneity is driven by the relative strength of the competitive and online steering effects: older and wealthier consumers (group 6) gain substantially regardless of their willingness to purchase online, while poorer groups face meaningful losses driven by the online steering effect raising their in-person prices.

Profits. Panel B of Table 10 reports the consequences of the online channel for industry profits. Under limited access with a moderate reduction in search costs, aggregate profits slightly decrease by €1.5 million, as the competitive effect compresses margins faster

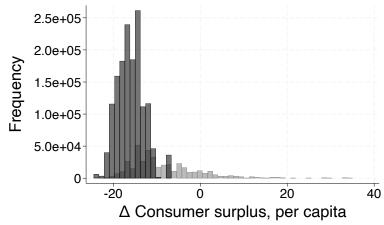
Figure 8: Change in per capita consumer surplus (-100% search cost online)



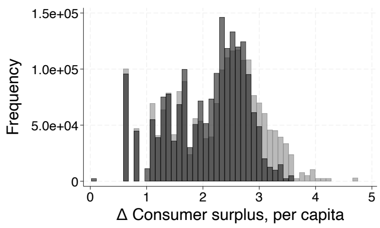
(a) Young/Poor



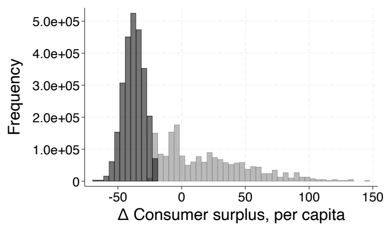
(b) Middle/Poor



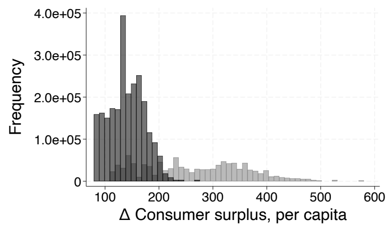
(c) Old/Poor



(d) Young/Rich



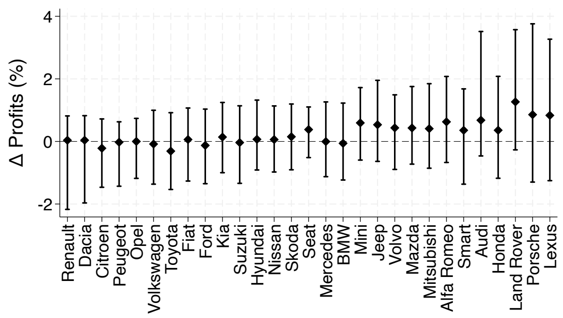
(e) Middle/Rich



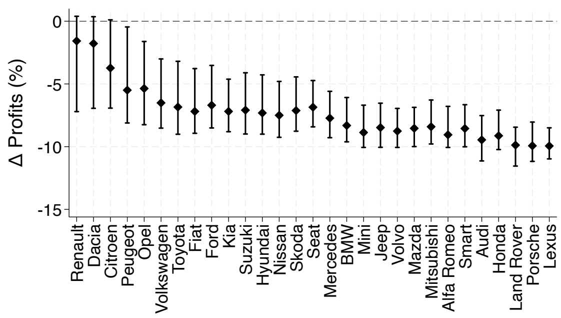
(f) Old/Rich

Notes: These figures plot the change in consumer surplus by demographic group due to the introduction of an online channel with limited access and no search costs, as per Table 8, column (3). Consumer surplus is the average per capita consumer surplus at the level of the municipality, and its distribution is weighted by group-specific populations.

Figure 9: Change in brand-level profits (-100% search cost online)



(a) Online plus in-person sales



(b) In-person sales only

Notes: These figures illustrate the change in profits at the brand level due to the introduction of an online channel with limited access and no search costs, as per Table 8, column (3). The marker represents the median and the brackets represent the 10th-90th percentile range. Brands are ordered on the x-axis by the total number of car dealers, in decreasing order.

than the market expands. In all other scenarios, profits increase, reaching +€136.8 million (+3.3%) under universal access with full elimination of search costs, where the gains from market expansion dominate.

Figure 9 presents the distribution of profit changes at the brand level for the scenario with full elimination of search costs and limited access to the online channel. Each marker represents the median profit change across all dealers selling a given brand, and the brackets indicate the 10th–90th percentile range. Brands are ordered according to the size of their dealer network, with the most accessible brands on the left.

Panel (a) of Figure 9 reveals two important features. First, for brands with large dealer networks, the median dealer experiences essentially no gain or loss in profits, with winners and losers roughly balancing out. Second, for brands with smaller and sparser dealer networks, the entire distribution shifts upward—the median store experiences an increase in profits, reaching around 1.3% for the most geographically remote brands. This pattern directly reflects the mechanism at work: eliminating search costs disproportionately benefits brands whose dealers are harder to reach, as consumers can now access their models without incurring high search costs.

Panel (b) of Figure 9 focuses solely on the profits from the in-person channel. As expected, the introduction of an online channel diverts a large portion of sales from car dealers, resulting in losses for the in-person channel ranging from 1.6% to 10% for the median dealer within a brand. These findings raise several questions about the future of car dealers. We investigate some of these questions next.

7.5 Exit of car dealers and cost efficiencies

To conclude our analysis, we assess the welfare consequences of two of our maintained assumptions. The first is that the online channel does not lead to exit of car dealers, while the second is that it does not lead to any savings on marginal costs.

A plausible consequence of the introduction of an online distribution channel is to drive out of business some car dealers. If this were the case, our estimates could overstate the associated welfare gains. Since we do not model entry decisions explicitly, we proceed by closing a certain number of car dealers and re-evaluating welfare in this new environment. We base this investigation on a scenario with limited access to the online channel for some consumers and $\rho = 1$ (as in Table 8, column (3)). We consider three counterfactuals: closing the 5%, 10%, and 20% least profitable car dealers, respectively. Detailed counterfactual results are reported in Appendix Table A.6, while we summarize the implied welfare changes in Table 11.

These counterfactuals reveal that the exit of car dealers can meaningfully affect welfare. We focus on the most extreme scenario where 20% of car dealers go out of business

Table 11: Exit of car dealers and cost efficiencies

	Exit of car dealers			Cost efficiencies	
	-5%	-10%	-20%	-5%	-10%
Panel A: Change in consumer surplus, in € per consumer					
Group 1: Young/Poor	0.0	-0.5	-1.8	+0.3	+0.3
Group 2: Young/Rich	+0.4	+0.3	-0.1	+0.5	+0.5
Group 3: Middle/Poor	+0.6	-1.1	-6.5	+1.8	+1.8
Group 4: Middle/Rich	-0.5	-1.1	-4.3	+0.1	+0.1
Group 5: Old/Poor	-3.3	-5.4	-12.4	-1.9	-1.9
Group 6: Old/Rich	+0.5	+0.7	+3.2	+0.4	+0.4
All consumers	-0.2	-0.8	-2.8	+0.3	+0.3
Panel B: Change in profits					
Profits, in million €	-2.9	-6.7	-18.8	+116.3	+305.3
Profits, in %	-0.07	-0.16	-0.46	+2.85	+7.47

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is a scenario with limited access to the online channel and no search costs, as per Table 8, column (3). All columns report welfare differences with respect to this baseline. In the first set of counterfactuals, we close 5%, 10%, or 20% of the least profitable car dealers. In the second set, we reduce marginal costs by 5% or 10% and impose a €400 delivery cost on online sales. All monetary values are in 2018 euros. Profits are in million euros.

after the introduction of the online channel. When no car dealer exits the market, the average gain in consumer surplus is around €27.5 (per consumer-year) and industry profits increase by €24.4 million (Table 10). When instead 20% of car dealers go out of business, the average gain in consumer surplus is lower by €2.8 per consumer-year. The increase in industry profits is also reduced by €18.8 million in this worst-case scenario, reversing most of the profit gains from operating the online channel. For a more realistic reallocation with 10% exit of car dealers, the reduction in consumer surplus is €0.8 per consumer-year and the additional decrease in industry profits is €6.7 million.

Another plausible consequence of the introduction of an online channel is that the marginal costs of manufacturers changed. For example, they could save on the costs associated with the "middleman," such as double marginalization and the micro-management of car dealers' inventories. We implement this by assuming that car manufacturers save some costs when selling directly to consumers through the online channel. Since we do not model vertical relations explicitly, we assume that selling directly to consumers entails a small reduction in marginal costs, in the range of 5-10%.[26]

We also assume that car manufacturers must incur a cost for delivering their vehicles to

[26] We take this range of marginal cost reductions from the analysis of vertical relations in the European car market by Brenkers and Verboven (2006), who estimate it to be around 7–8%. Since our estimates suggest an average marginal cost of €18,427, a 5% (10%) reduction corresponds to around €900 (€1800).

the consumers' doorsteps. To evaluate these delivery costs, we use an online platform specialized in car deliveries, Shiply.com,[27] and ask quotes for various vehicle deliveries for a selection of city pairs in France (shortest distance inquired: 75km, longest distance inquired: 250km). Since all quotes are between €300 and €500 for a single car delivery, we choose an average of €400, which amounts to around 2% of the estimated average marginal cost (see footnote 26). Detailed counterfactual results are reported in Appendix Table A.6, while we report the welfare changes in columns (4) and (5) of Table 11.

In line with intuition, if the online channel also allowed car manufacturers to save on intermediate costs (such as double marginalization and extra inventories), equilibrium prices would decrease, overall car sales would increase and, as a consequence, both consumer surplus and industry profits would substantially increase. In relation to this scenario, our baseline specification that assumes the same marginal costs for both distribution channels would underestimate the overall benefits of the introduction of an online channel. Importantly, the fact that *industry* profit would substantially increase (by €116.3 million, or +2.85%, in the case of a 5% marginal cost reduction, and by €305.3 million, or +7.47%, for a 10% reduction) means that, in theory, there could be ways of redistributing aggregate profit so as to guarantee that car dealers would remain as well off as in the scenario without an online channel (or even better off). That is, car dealers could be asked not to charge any margin on car sales, be more than fully compensated with lump-sum transfers, and car manufacturers would still make more profit than in the absence of an online channel.

8 Conclusion

We investigate the welfare consequences of introducing an online distribution channel in the French car industry. We propose a structural model of oligopolistic competition with differentiated products that combines unobserved third-degree price discrimination and sequential consumer search. We estimate this model using data from 2009–2021, a period in which virtually all car sales took place in person (with the exception of Tesla from 2019 onward). For our counterfactual analysis, we extend the model to multi-channel sequential search: consumers choose, for each car manufacturer, whether to search in person or online, and then search sequentially across manufacturers. Online purchases take place at list prices with reduced search costs, while in-person purchases allow for price discrimination but involve full search costs.

We document that price discrimination plays a less prominent role than search costs

[27] Source: https://www.shiply.com.

in this industry. Eliminating search costs yields welfare benefits around four times as large as those of price discrimination for consumers and twice as large for firms. This suggests that the profitability of an online distribution channel hinges on its ability to reduce search costs substantially.

When the online channel is introduced, two forces shape the new pricing equilibrium. The *competitive effect* intensifies price competition by allowing consumers to access car models from manufacturers whose dealers were previously too distant, driving list prices down. The *online steering effect* leads firms to raise in-person prices for some consumer groups, steering them toward the more profitable online channel. Firms generally continue to offer in-person discounts for the most price-sensitive consumers (younger and less wealthy), but compress or eliminate discounts for other groups. When access to the online channel is universal and search costs are fully eliminated, transaction prices converge across channels for most consumer groups.

In terms of welfare, the online channel leads to market expansion and is generally profitable for car manufacturers, with industry profit increasing by up to 3.3%. Average consumer surplus also increases, but the gains are unevenly distributed. Older and wealthier consumers—who would typically not receive in-person discounts—benefit the most from both lower list prices and search cost savings. Less wealthy consumers in middle-aged and older groups, whose in-person prices rise due to the online steering effect, can experience meaningful losses. Younger consumers are less affected, as their large baseline discounts insulate them from the online channel. The distributional consequences of the online channel depend on the magnitude of search cost reductions and on the share of consumers willing to purchase online.

We also find that expanding consumer access to the online channel has a larger effect on online penetration than further reducing search costs, and that the introduction of an online channel by a single manufacturer exerts competitive pressure on rivals' prices even when few consumers switch firms.

Our analysis is subject to two important caveats. First, we assume that the online distribution channel does not alter car dealer networks. Second, we assume that selling directly to consumers online does not lead to savings on marginal costs, which could arise when bypassing the "middleman." Although we probe the robustness of our results along both dimensions, our model is not equipped to fully address these mechanisms, and we leave a thorough investigation to future research.

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Appendix

A Proof of $\mathbb{E}[\mathbf{sc}_{ifdm}] = \overline{\mathbf{sc}}^{f}_{idm}$

Here we show that if the search cost $\mathrm{sc}_{ifdm} \geq 0$ is distributed according to the c.d.f. G^{sc}_{ifdm} defined in (5) with $\overline{\mathrm{sc}}^{f}_{idm} \geq 0$, then $\mathbb{E}[\mathrm{sc}_{ifdm}] = \overline{\mathrm{sc}}^{f}_{idm}$.

Let $S(x) \equiv 1 - \exp(-\exp(-x))$ denote the survival function of the standard Gumbel distribution. For $z > 0$, the c.d.f. in (5) can be written as:

$$G^{\mathrm{sc}}_{ifdm}(z) = \frac{S\left(H_0^{-1}(z) + \overline{\mathrm{sc}}^{f}_{idm}\right)}{S\left(H_0^{-1}(z)\right)}. \tag{30}$$

From (5), the distribution of sc_{ifdm} has a point mass at zero of magnitude:

$$\Pr(\mathrm{sc}_{ifdm} = 0) = G^{\mathrm{sc}}_{ifdm}(0) - \lim_{z \to 0^-} G^{\mathrm{sc}}_{ifdm}(z) = \exp(-\overline{\mathrm{sc}}^{f}_{idm}) - 0 = \exp(-\overline{\mathrm{sc}}^{f}_{idm}), \tag{31}$$

with the remaining mass $1 - \exp(-\overline{\mathrm{sc}}^{f}_{idm})$ distributed continuously on $(0, \infty)$. By definition, the expectation of sc_{ifdm} is:

$$\mathbb{E}[\mathrm{sc}_{ifdm}] = \int_{[0,\infty)} z\, dG^{\mathrm{sc}}_{ifdm}(z) = 0 \cdot \Pr(\mathrm{sc}_{ifdm} = 0) + \int_{(0,\infty)} z\, dG^{\mathrm{sc}}_{ifdm}(z) = \int_{(0,\infty)} z\, dG^{\mathrm{sc}}_{ifdm}(z), \tag{32}$$

where the point mass contributes zero since it is located at $z = 0$. The tail integral formula for non-negative random variables then gives:

$$\int_{(0,\infty)} z\, dG^{\mathrm{sc}}_{ifdm}(z) = \int_0^\infty \Pr(\mathrm{sc}_{ifdm} > z)\, dz = \int_0^\infty \left[1 - G^{\mathrm{sc}}_{ifdm}(z)\right] dz, \tag{33}$$

where, for $z > 0$:

$$1 - G^{\mathrm{sc}}_{ifdm}(z) = \frac{S\left(H_0^{-1}(z)\right) - S\left(H_0^{-1}(z) + \overline{\mathrm{sc}}^{f}_{idm}\right)}{S\left(H_0^{-1}(z)\right)}. \tag{34}$$

We apply the change of variables $u = H_0^{-1}(z)$, or equivalently $z = H_0(u)$. Recall that H_0^{-1} is the inverse of $H_0(r) = \int_r^\infty (z - r)\, dG(z)$, which can equivalently be expressed as $H_0(r) = \int_r^\infty S(t)\, dt$ using integration by parts. Then, since $H_0'(u) = -S(u)$, we have $dz = -S(u)\, du$. As z ranges from 0 to ∞, u ranges from $+\infty$ to $-\infty$. Substituting into

(33) we then obtain:

$$\begin{aligned}\mathbb{E}[\mathrm{sc}_{ifdm}] &= \int_{+\infty}^{-\infty} \frac{S(u) - S\left(u + \overline{\mathrm{sc}}_{idm}^{f}\right)}{S(u)} \cdot (-S(u))\, du \\ &= \int_{-\infty}^{+\infty} \left[S(u) - S\left(u + \overline{\mathrm{sc}}_{idm}^{f}\right)\right] du. \end{aligned} \tag{35}$$

To evaluate (35), consider the integral over a finite interval $[a, b]$:

$$\begin{aligned}\int_{a}^{b} \left[S(u) - S\left(u + \overline{\mathrm{sc}}_{idm}^{f}\right)\right] du &= \int_{a}^{b} S(u)\, du - \int_{a+\overline{\mathrm{sc}}_{idm}^{f}}^{b+\overline{\mathrm{sc}}_{idm}^{f}} S(v)\, dv \\ &= \int_{a}^{a+\overline{\mathrm{sc}}_{idm}^{f}} S(u)\, du - \int_{b}^{b+\overline{\mathrm{sc}}_{idm}^{f}} S(u)\, du. \end{aligned} \tag{36}$$

Taking limits in (36): as $a \to -\infty$, $S(u) \to 1$ uniformly on $[a, a + \overline{\mathrm{sc}}_{idm}^{f}]$, so the first integral converges to $\overline{\mathrm{sc}}_{idm}^{f}$; as $b \to +\infty$, $S(u) \to 0$ uniformly on $[b, b + \overline{\mathrm{sc}}_{idm}^{f}]$, so the second integral converges to 0. Consequently, $\mathbb{E}[\mathrm{sc}_{ifdm}] = \overline{\mathrm{sc}}_{idm}^{f}$.

B Additional Tables and Figures

Table A.1: Demographic characteristics by group

Description	Mean	Std. dev.	10th pct.	Median	90th pct.	Observations
Group 1: Age < 40, *Income = Low*						
Median income	16,680	2,688	12,973	17,122	19,760	135,647
Average age	26.9	0.9	25.8	26.8	27.9	135,647
Share of female	0.500	0.030	0.471	0.503	0.523	135,647
Average household size	2.22	0.27	1.92	2.16	2.58	135,647
Urban	0.474	0.499	0	0	1	135,647
Share of population (ϕ_1)	0.245					13
Group 2: Age < 40, *Income = High*						
Median income	24,831	4,237	20,827	23,366	31,585	147,467
Average age	27.5	1.0	26.4	27.5	28.6	147,467
Share of female	0.503	0.036	0.463	0.509	0.535	147,467
Average household size	2.35	0.26	1.98	2.38	2.67	147,467
Urban	0.465	0.499	0.000	0.000	1.000	147,467
Share of population (ϕ_2)	0.142					13
Group 3: Age $\in [40, 60)$, *Income = Low*						
Median income	17,880	2,259	14,980	18,385	20,231	161,066
Average age	49.5	0.7	48.9	49.5	50.2	161,066
Share of female	0.511	0.034	0.472	0.517	0.540	161,066
Average household size	2.23	0.27	1.92	2.18	2.59	161,066
Urban	0.373	0.484	0	0	1	161,066
Share of population (ϕ_3)	0.151					13
Group 4: Age $\in [40, 60)$, *Income = High*						
Median income	25,220	5,122	20,853	23,659	31,800	190,650
Average age	49.4	0.7	48.7	49.4	50.1	190,650
Share of female	0.511	0.030	0.475	0.516	0.539	190,650
Average household size	2.33	0.25	1.97	2.36	2.65	190,650
Urban	0.353	0.478	0	0	1	190,650
Share of population (ϕ_4)	0.188					13
Group 5: Age ≥ 60, *Income = Low*						
Median income	18,859	1,555	16,839	19,092	20,605	161,343
Average age	70.5	1.2	69.1	70.5	71.8	161,343
Share of female	0.547	0.048	0.488	0.553	0.596	161,343
Average household size	2.24	0.25	1.94	2.21	2.55	161,343
Urban	0.201	0.400	0	0	1	161,343
Share of population (ϕ_5)	0.093					13
Group 6: Age ≥ 60, *Income = High*						
Median income	24,648	4,650	20,934	23,313	29,777	190,053
Average age	70.1	1.0	68.9	70.2	71.2	190,053
Share of female	0.552	0.040	0.500	0.558	0.592	190,053
Average household size	2.25	0.26	1.92	2.23	2.59	190,053
Urban	0.400	0.490	0	0	1	190,053
Share of population (ϕ_6)	0.182					13

Notes: Statistics concerning the median income, age, household size, the share of women, and the level of urbanity are weighted by municipal-level group-specific populations. Statistics concerning the willingness to purchase cars online are weighted by survey weights. We report a simple year-over-year average of the group-specific population shares, denoted by ϕ_d.

Table A.2: Evidence of price dispersion

	Transaction price		Transaction price net of buyback value	
	(1)	(2)	(3)	(4)
Income	40.497***	25.916**	39.253***	28.970**
	(13.112)	(11.958)	(13.760)	(14.533)
Age	67.387***	20.156	47.276***	2.754
	(15.828)	(19.359)	(17.284)	(25.626)
Female	851.802	141.661	1,051.810	604.796
	(1,078.583)	(1,463.399)	(1,147.382)	(1,893.736)
Age × Female	−17.770	−5.874	−25.134	−13.574
	(19.998)	(26.556)	(21.727)	(32.152)
Value of down payment	0.006	0.005	0.013	0.011**
	(0.005)	(0.003)	(0.008)	(0.005)
Household: 2 pers.	−182.571	−250.977	−478.245	−707.159
	(393.500)	(633.025)	(500.774)	(815.847)
Household: 3 pers.	−688.958	−786.539	−954.493	−1,117.172
	(620.113)	(925.880)	(709.379)	(1,208.438)
Household: 4 pers.	−644.082	−1,357.238*	−1,464.877**	−2,485.373**
	(626.979)	(802.681)	(655.590)	(1,044.736)
Household: 5 pers.	−3,000.296***	−2,397.637**	−2,987.593***	−4,892.123***
	(882.021)	(1,109.743)	(962.957)	(1,589.441)
Household: 6+ pers.	−202.262	−929.100	1,490.901	−399.893
	(2,294.165)	(2,114.791)	(2,113.088)	(2,009.281)
Urban area: less than 15,000	−825.524	2,275.435	−1,235.378	2,558.358
	(1,425.966)	(1,552.297)	(2,290.406)	(3,842.866)
Urban area: 15,000–24,999	345.570	548.107	1,610.750	2,790.948
	(1,717.314)	(1,827.368)	(1,458.979)	(2,818.982)
Urban area: 25,000–34,999	−1,588.274	1,024.648	−1,243.062	1,283.947
	(1,377.100)	(2,077.463)	(1,659.565)	(3,108.214)
Urban area: 35,000–49,999	−1,733.160	−6.787	−1,418.662	1,805.762
	(1,095.743)	(1,002.154)	(1,220.085)	(1,598.266)
Urban area: 50,000–99,999	−1,316.561	−462.050	−1,999.194**	−1,382.434
	(815.740)	(1,136.611)	(785.564)	(1,128.750)
Urban area: 100,000–199,999	−822.825	195.033	−198.663	222.065
	(791.007)	(930.849)	(714.154)	(1,036.763)
Urban area: 200,000–499,999	−1,093.716	182.223	−864.898	−6.343
	(697.664)	(876.896)	(608.700)	(1,023.145)
Urban area: 500,000 or more	−1,143.714*	−366.690	−795.605	197.814
	(657.948)	(888.946)	(643.131)	(882.634)
Urban area: Paris greater metro area	−900.305	−523.017	−121.970	189.700
	(731.188)	(1,140.885)	(676.618)	(990.022)
New vehicles only	No	Yes	No	Yes
Fixed effects				
Car model × engine × new/used	Yes	Yes	Yes	Yes
Year × month of purchase	Yes	Yes	Yes	Yes
Country of origin of buyer	Yes	Yes	Yes	Yes
Observations	1,283	698	1,283	698
R-squared	0.742	0.795	0.600	0.617

Notes: This table presents the result of a regression of transaction prices on demographic characteristics of buyers, based on a survey of consumers' expenditures. Columns (1) and (3) include sales of both new and used cars, purchased at a car dealer. Columns (2) and (4) include only new cars. The buyback value represents the payment that was received by the consumer for trading in their old car. Standard errors in parentheses are clustered at the car model × engine × new/used level. Significance: * < 0.10, ** < 0.05, *** < 0.01.

Table A.3: Evidence of price dispersion, by demographic group

	Transaction price		Transaction price – buyback value	
	(1)	(2)	(3)	(4)
Group 1: Age < 40, Income $=$ Low	— Base category (omitted) —			
Group 2: Age < 40, Income $=$ High	1,891.780**	1,672.480*	2,268.324***	2,303.590*
	(744.064)	(910.848)	(710.450)	(1,215.509)
Group 3: Age $[40, 60)$, Income $=$ Low	2,174.785***	1,891.699**	2,766.616***	1,472.772
	(611.006)	(899.649)	(708.275)	(1,307.562)
Group 4: Age $[40, 60)$, Income $=$ High	3,734.329***	3,482.132**	2,961.747***	1,845.544
	(826.167)	(1,418.426)	(853.483)	(1,763.667)
Group 5: Age $\geq g0$, Income $=$ Low	2,412.376***	1,843.251*	1,498.397**	80.061
	(640.156)	(959.263)	(690.785)	(1,148.006)
Group 6: Age ≥ 60, Income $=$ High	2,861.136***	811.124	1,725.900**	-477.457
	(773.938)	(832.586)	(803.734)	(1,130.036)
Female	-306.879	-571.473	-510.496	-516.426
	(346.764)	(487.459)	(372.094)	(621.099)
Value of down payment	0.007	0.006*	0.012	0.011**
	(0.005)	(0.003)	(0.008)	(0.005)
Household: 2 pers.	63.960	-297.050	-133.333	-561.202
	(387.543)	(598.280)	(479.113)	(820.017)
Household: 3 pers.	-151.740	-942.284	-551.323	-1,268.167
	(602.009)	(815.275)	(660.687)	(1,140.575)
Household: 4 pers.	117.470	-1,524.256*	-1,019.702*	-2,687.011**
	(548.932)	(785.119)	(550.327)	(1,132.520)
Household: 5 pers.	-2,166.131***	-1,871.581	-2,437.147***	-4,361.964***
	(794.271)	(1,161.155)	(874.917)	(1,666.250)
Household: 6+ pers.	1,709.236	-1,236.283	3,163.307	-315.518
	(3,021.293)	(1,943.145)	(2,911.596)	(1,961.040)
Urban area: less than 15,000	-1,016.912	2,739.753*	-1,576.286	3,032.271
	(1,326.101)	(1,429.939)	(2,297.180)	(3,751.639)
Urban area: 15,000–24,999	288.247	1,084.762	1,611.884	2,909.341
	(1,581.361)	(1,614.560)	(1,441.358)	(2,866.254)
Urban area: 25,000–34,999	-1,493.065	1,596.119	-1,166.079	1,902.540
	(1,370.082)	(1,903.312)	(1,654.036)	(2,956.974)
Urban area: 35,000–49,999	-1,823.739	-132.945	-1,590.025	1,627.714
	(1,187.328)	(975.182)	(1,281.310)	(1,626.837)
Urban area: 50,000–99,999	-1,339.540	-528.954	-2,156.797***	-1,535.573
	(837.413)	(1,113.736)	(795.857)	(1,116.164)
Urban area: 100,000–199,999	-760.895	316.345	-196.287	325.665
	(803.541)	(876.063)	(699.684)	(998.607)
Urban area: 200,000–499,999	-943.257	215.036	-700.908	84.523
	(713.562)	(835.990)	(617.439)	(1,019.786)
Urban area: 500,000 or more	-971.024	-55.535	-626.253	373.406
	(642.346)	(828.815)	(620.582)	(873.457)
Urban area: Paris greater metro area	-583.891	-67.373	280.185	748.892
	(719.717)	(991.942)	(661.801)	(919.323)
New vehicles only	No	Yes	No	Yes
Fixed effects				
Car model $\times$ engine $\times$ new	Yes	Yes	Yes	Yes
Year of purchase $\times$ month	Yes	Yes	Yes	Yes
Country of origin of buyer	Yes	Yes	Yes	Yes
Fstat	5.41	1.47	3.96	1.61
Pr(Fstat) $>$ F	< 0.001	0.205	0.002	0.164
Observations	1,283	698	1,283	698
R-squared	0.740	0.801	0.600	0.620

Notes: This table presents the result of a regression of transaction prices on demographic group indicators and other demographic characteristics of buyers, based on a survey of consumers' expenditure. We have excluded observations where the car was purchased following an insurance claim (i.e., the replacement of a damaged vehicle). Columns (1) and (3) include sales of both new and used cars, purchased at a car dealer. Columns (2) and (4) include only new car purchases. The buyback value represents the payment that was received by the consumer for trading in their old car. The F-statistic tests for the hypothesis that the coefficients on the group indicators are jointly zero. Standard errors in parentheses are clustered at the car model $\times$ engine $\times$ new/used level. Significance: * < 0.10, ** < 0.05, *** < 0.01.

Table A.4: Car dealers' market presence, by brand

Brand	Stores	Market share	Distance to consumers, in km				
		(%)	Mean	Std. dev.	10th pct.	Median	90th pct.
Renault	1132	17.8	8.85	8.82	1.98	5.19	21.48
Dacia	1101	8.5	9.01	9.01	2.02	5.27	21.77
Peugeot	401	15.9	18.50	16.16	3.43	13.13	42.16
Citroen	396	12.5	13.31	12.09	3.32	8.17	30.47
Opel	351	3.6	14.72	13.61	3.64	8.89	33.83
Volkswagen	303	7.2	15.66	14.54	3.09	9.53	36.88
Toyota	264	5.2	16.27	14.86	3.41	9.76	37.75
Fiat	260	2.7	17.18	15.65	3.57	10.58	40.77
Ford	252	4.6	17.48	15.63	4.04	10.62	40.29
Jeep	225	0.2	28.88	26.67	4.52	18.68	69.56
Kia	215	2.3	17.98	16.47	3.84	10.73	42.42
Suzuki	212	1.5	18.01	16.25	3.80	11.12	42.10
Alfa Romeo	208	0.4	29.63	27.69	4.83	19.18	71.17
Hyundai	198	1.7	18.74	17.09	3.89	11.37	42.80
Nissan	191	2.9	18.90	16.79	3.88	11.79	44.05
Skoda	186	1.1	20.38	18.37	4.02	12.74	48.31
Seat	157	1.6	21.33	18.84	4.56	14.10	48.50
BMW	152	1.8	22.43	20.77	4.16	13.94	53.21
Mercedes	152	2.0	22.63	19.70	4.08	16.03	52.43
Mini	133	1.4	24.79	23.16	4.19	15.82	59.38
Volvo	116	0.5	26.23	24.15	4.45	16.91	62.72
Mazda	112	0.5	28.85	27.36	5.30	18.61	67.36
Mitsubishi	109	0.2	30.28	28.08	5.50	18.97	72.27
Smart	102	0.2	32.58	30.87	4.49	20.29	79.77
Audi	88	2.5	45.68	40.24	7.16	33.56	99.71
Honda	76	0.5	36.59	33.12	5.98	24.05	85.9
Land Rover	62	0.3	43.65	40.81	5.34	29.70	106.2
Porsche	45	0.1	56.43	49.90	6.32	42.68	132.0
Lexus	42	0.2	54.2	48.63	5.92	41.71	127.4
Tesla	16	0.2	96.1	76.92	9.36	79.94	209.0
TOTAL	7,241	100					

Notes: Brands are ordered by their number of dealers. The market share is computed as each brand's sales over total sales. Driving distance to consumers is computed over all demographic groups, weighted by their respective municipal-level populations.

Table A.5: Car sales, distances to dealers, and car characteristics

Description	Mean	Std. dev.	10th pct.	Median	90th pct.	Observations
Panel A: Sales, by product-year						
Group 1: Young/Poor	285	654	10	67	740	4,960
Group 2: Young/Rich	244	564	10	61	599	4,960
Group 3: Middle/Poor	395	851	22	112	972	4,960
Group 4: Middle/Rich	711	1,467	47	214	1,791	4,960
Group 5: Old/Poor	283	740	10	55	651	4,960
Group 6: Old/Rich	748	1,778	33	176	1,718	4,960
Panel B: Distance, km						
Group 1: Young/Poor	16.2	19.0	2.9	8.2	39.3	613,080
Group 2: Young/Rich	14.4	16.5	2.5	9.1	31.7	573,660
Group 3: Middle/Poor	18.8	20.1	3.0	10.2	44.8	594,210
Group 4: Middle/Rich	15.5	17.9	2.9	9.4	34.3	620,220
Group 5: Old/Poor	21.0	19.0	3.1	16.1	46.1	588,780
Group 6: Old/Rich	14.7	16.8	2.8	8.3	34.2	627,840
Panel C: Car characteristics						
Net list price, in €	22,742	9,737	12,718	20,527	33,808	4,960
Horsepower, in kW	75.7	22.9	51.0	70.0	103.0	4,960
Weight, in kg	1,737	274	1,418	1,700	2,080	4,960
Fuel cost, in €/100km	6.5	1.6	4.7	6.5	8.4	4,960
Fuel consumption, in L/100km	4.7	1.0	3.7	4.7	5.9	4,960
Gasoline	0.49	0.50	0	0	1	4,960
Diesel	0.45	0.50	0	0	1	4,960
Electric	0.02	0.13	0	0	0	4,960
Plug-in hybrid	0.00	0.07	0	0	0	4,960
Hybrid	0.04	0.19	0	0	0	4,960
Sedan	0.73	0.44	0	1	1	4,960
Convertible	0.01	0.08	0	0	0	4,960
Station wagon	0.26	0.44	0	0	1	4,960

Notes: Panel A reports mean sales by product, year, and demographic group (unweighted). Panel B reports driving distances to the closest dealer of each brand, weighted by brand-level national sales and group-specific municipal populations. Panel C reports car characteristics weighted by aggregate sales. All monetary values are in 2018 euros.

Table A.6: Exit of car dealers and cost efficiencies (detailed results)

	Baseline	Exit of car dealers			Cost efficiencies	
		-5%	-10%	-20%	-5%	-10%
Panel A: Transaction prices, uniform weights						
Group 1: Young/Poor	21,427	21,424	21,426	21,433	21,422	21,422
Group 2: Young/Rich	21,645	21,640	21,640	21,641	21,640	21,640
Group 3: Middle/Poor	22,300	22,289	22,294	22,309	22,286	22,286
Group 4: Middle/Rich	22,916	22,913	22,914	22,920	22,912	22,912
Group 5: Old/Poor	22,205	22,217	22,221	22,233	22,214	22,214
Group 6: Old/Rich	23,217	23,212	23,207	23,182	23,217	23,217
Online/List price	23,217	23,212	23,207	23,182	23,217	23,217
Panel B: Transaction prices, sales-weighted						
Group 1: Young/Poor	20,098	20,084	20,066	20,021	20,094	20,094
Group 2: Young/Rich	21,913	22,019	22,016	21,999	22,025	22,025
Group 3: Middle/Poor	21,357	21,374	21,359	21,341	21,386	21,386
Group 4: Middle/Rich	23,745	23,771	23,770	23,803	23,768	23,768
Group 5: Old/Poor	20,403	20,435	20,422	20,402	20,445	20,445
Group 6: Old/Rich	23,205	23,193	23,179	23,127	23,205	23,205
Online/List price	29,570	29,307	29,229	28,945	29,324	29,324
Panel C: Sales, in units						
Group 1: Young/Poor	108,701	-3,274	-3,474	-4,095	-3,138	-3,138
Group 2: Young/Rich	78,774	-8,036	-8,063	-8,152	-8,016	-8,016
Group 3: Middle/Poor	137,170	-1,164	-1,516	-2,616	-924	-924
Group 4: Middle/Rich	234,792	-6,888	-6,999	-7,570	-6,784	-6,784
Group 5: Old/Poor	110,275	+1,126	+854	-28	+1,297	+1,297
Group 6: Old/Rich	335,002	-6,215	-6,200	-5,840	-6,225	-6,225
All consumers	1,004,714	-24,451	-25,398	-28,301	-23,790	-23,790
Panel D: Proportion of online sales						
Group 1: Young/Poor	0.010	0.009	0.009	0.011	0.008	0.008
Group 2: Young/Rich	0.010	0.009	0.009	0.010	0.009	0.009
Group 3: Middle/Poor	0.116	0.111	0.115	0.125	0.109	0.109
Group 4: Middle/Rich	0.223	0.222	0.224	0.234	0.220	0.220
Group 5: Old/Poor	0.036	0.039	0.041	0.046	0.038	0.038
Group 6: Old/Rich	0.216	0.213	0.214	0.216	0.212	0.212
Panel E: Average distance to car models, in km						
Group 1: Young/Poor	14.78	+1.07	+1.41	+2.40	+0.81	+0.81
Group 2: Young/Rich	12.36	+2.04	+2.23	+2.79	+1.90	+1.90
Group 3: Middle/Poor	18.47	+0.35	+0.82	+2.16	+0.02	+0.02
Group 4: Middle/Rich	15.47	+0.35	+0.64	+1.57	+0.16	+0.16
Group 5: Old/Poor	22.19	-0.38	+0.06	+1.41	-0.63	-0.63
Group 6: Old/Rich	14.26	+0.29	+0.58	+1.46	+0.06	+0.06

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is a scenario with limited access to the online channel and no search costs, as per Table 8, column (3). All monetary values are in 2018 euros. Uniform-weighted transaction prices (Panel A) use weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups and counterfactual experiments. Sales-weighted transaction prices (Panel B) use realized sales for each demographic group and counterfactual experiment. For sales and average distances, we report values at baseline in the first column and changes from baseline in the other columns. In the first set of counterfactuals, we close 5%, 10%, or 20% of the least profitable car dealers. In the second set, we reduce marginal costs by 5% or 10% and impose a €400 delivery cost on online sales.

C Computational details

In this section, we provide additional computational details related to the data, the two-step estimation routine, and the counterfactual simulations.

C.1 Additional details on the data

Construction of demographic groups. We provide more details on the construction of demographic groups. We collect data from two sources, a population survey by municipality and age group, available every five years, and an income survey by municipality and age group, available yearly. Both datasets are available from the National Institute of Statistics and Economic Studies (INSEE).[28] We use the following age categorization: young (39 or younger), middle aged (between 40 and 59 included) and old (60 or older). Within age category, we divide municipalities into two evenly sized groups, high- and low-income, according to the median income reported in the income files. Since income is reported for finer increments in age than our age categories, we use a population-weighted average of the median income within age groups and municipalities to assign an income group. In some cases, for very small municipalities, income is not reported separately by age. We then assign the median income of the municipality to all age groups. We drop a small number of municipalities that are too small to report income at all (along with the associated car sales). We obtain six demographic groups, described in Table A.7 below.

Table A.7: Demographic groups definition

Group	Definition	
Group 1	Age 39 or younger	Income in bottom half of age-specific distribution
Group 2	Age 39 or younger	Income in top half of age-specific income distribution
Group 3	Age between 40 and 59	Income in bottom half of age-specific distribution
Group 4	Age between 40 and 59	Income in top half of age-specific distribution
Group 5	Age 60 or older	Income in bottom half of age-specific distribution
Group 6	Age 60 or older	Income in top half of age-specific distribution

Demographic characteristics. We use the population census and income files described above to construct the demographic group-by-municipality average characteristics of consumers. A summary of these consumer characteristics by demographic group is available in Table A.1.

The population files include both the population by municipality and age and the number

[28] Source: https://www.insee.fr/.

of households. We use the number of households (divided by 4) to define market size, and we compute the average household size using the ratio of population to the number of households. The median income is computed as described above and the average age can be approximated with the population data, taking the midpoint of age intervals (5-year increments) and using population weights. Although income and age are used to define the groups, we find that the within-group variation (at the demographic group-by-municipality level) is informative of consumers preferences. The population files offer a breakdown of populations by municipality, gender, and age, allowing us also to compute the share of women by demographic group and municipality.

We merge these data to a survey of population densities, also available at INSEE. Population density is available at the municipality level as a categorical variable indicating whether a given municipality is urban, suburban, or rural. Since not all demographic groups can be found in all municipalities, we define the indicator variable "urbanity" at the demographic group-by-municipality level. Finally, we use a survey of attitudes towards online shopping to determine the propensity to shop online, by demographic group (see Section 7.1 for details).

Construction of the car data. Our car data come from AAA data, which collects data on all car registrations in France. We obtain all new car registrations between 2009 and 2021. The data are aggregated at the level of the car model (a product), age group (in increments of 5 years), and municipality. We merge these data (using the age and the municipality of residence) to our data on consumer demographics to recover demographic groups based on the age and income of buyers, as described above. We create two main datasets. The first is aggregated at the level of the brand-model-engine-year-demographic group; this is our aggregated dataset used for the estimation of the linear parameters. The second dataset is aggregated at the level of the brand-model-engine-year-demographic group-municipality; this is our disaggregated dataset used to compute micro moments and estimate the nonlinear parameters.

We keep the 29 most prominent brands, and keep products with a net price (adjusted for the French Feebate Program) below €100,000. The car data include list prices and some common car characteristics such as horsepower. Horsepower and fuel consumption are not available for electric vehicles in the data. We impute the missing horsepower using an alternative data source and set the fuel consumption of electric vehicles to their fuel-equivalent electricity consumption. We compute fuel costs using various fuel prices interacted with fuel consumption, depending on the engine type (e.g., diesel prices for diesel engines). Finally, we obtain each vehicle's marketing segment (e.g., compact, SUV, etc.) and the country of origin of each model (e.g., the location of the plant that produces each model) from Jato Dynamics. Summary statistics about sales and the main car characteristics are presented in Table A.5.

We exclude Tesla from the set of manufacturers under consideration. There are two reasons for this. First, Tesla represents a very small share of total sales for several years of our data. Second, Tesla did not operate a physical network of dealers in France before 2020. There is also ample anecdotal evidence that Tesla does not price discriminate against consumers: consumers buy the car on the website at the posted price. Consequently, the inclusion of Tesla in the analysis would significantly complicate the model and its estimation, while only explaining a very small fraction of sales.

Construction of car dealership data. We obtain the location of all car dealers in France from manufacturers' websites in early 2024, for all brands under consideration except Smart, for which we were unable to scrape the website reliably. In that case, we obtained the addresses and names of car dealers from AutoConcession,[29] an online registry of French car dealers. This registry is not comprehensive for all brands: as an example, it contains about half of the Renault dealers we observe on Renault's website. For other brands, such as Mercedes-Benz, both datasets almost coincide. Since Smart is a brand associated with Mercedes-Benz, we assume that the data on AutoConcession are accurate for Smart also (several Mercedes-Benz dealers also sell Smart).

The data include the full name of dealers, by brand, their addresses, and the type of services offered (e.g., sale of new vehicles, sale of used vehicles, service, etc.). We remove all dealers that do not sell new vehicles. To compute the coordinates of each car dealer, we use OpenCage Geocoding API[30] to recover longitudes and latitudes from dealers' addresses. Most requests on OpenCage API returned a match at the street level or better, however, a small number returned a match at the postal code or city levels. We manually fix these low-quality matches using Google Maps.[31].

Our final dataset includes 7,241 car dealers, selling 29 brands. Summary statistics for these car dealers are available in Table A.4. We report on the number of dealers by brand, their combined market share by brand, and their proximity to consumers. Proximity is defined as the driving distance from consumers' location to the closest dealer of each brand. Additional details on the computation of these driving distances can be found in Section 4.2.

[29]Source: `https://www.autoconcession.fr`.
[30]`https://opencagedata.com`.
[31]Source: `https://www.google.com/maps/`.

C.2 Additional details on estimation

Notation and specification details. Table A.8 summarizes the notation used throughout the paper. Table A.9 provides additional details on the variables used in the model specification that we estimate. Unless indicated otherwise, car characteristics are used in both the demand specification and the marginal cost function. We include horsepower, curb weight, fuel cost or fuel consumption, and indicator variables for the engine type and body type. We take list prices as given and estimate transaction prices along with the model parameters. Driving distance to the closest dealer of product j is included in the demand specification and captures the search costs incurred by consumers to purchase product j. As emphasized in the main text, these search costs should be interpreted as inclusive of all interactions between the consumer and the car dealer, e.g., visiting the car dealer for a test drive and future visits related to service and maintenance. Demographic characteristics are used in the estimation of demand and for constructing micro moments. We include the (average) median income, the average age, the share of women, the average household size, and an indicator variable for urban municipalities. Intuitively, these demographic variables help us capture heterogeneous preferences across demographic groups and municipalities, while the random coefficients allow us to capture heterogeneity within each demographic group and municipality. Cost shifters enter the estimation of the marginal cost function and include an input price index and the real exchange rate between France and the country of origin of the vehicle, defined by the location of the manufacturing plant that produces each model. Both cost shifters are lagged by one year to reflect planning horizons. We also report on the demand- and supply-side (excluded) instruments used in the estimation, which are constructed following Berry et al. (1995). Additional details are available in Table A.9.

The model is estimated using a two-step GMM that estimates the nonlinear and the linear parameters sequentially. We find that this allows for greater computational tractability and a much faster estimation, since we can effectively separate the tasks of iterating over the nonlinear parameters (first step) and of solving for the unobserved transaction prices (second step). In what follows, we provide details on the estimation routine. To simplify notation, we omit the time subscript from the following expressions, even though estimation is performed over 13 years of data.

Estimation of the nonlinear parameters. Estimation of the nonlinear parameters $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ relies on the micro moments discussed in Sections 3 and 5. We provide an example using the distance to dealers, dist_{fm}, and the construction of the other micro moments follows the same logic. We specify the vector of micro moments $g_3(\Pi, \Sigma, \gamma, \tau)$

Table A.8: Model notation

Notation	Description
i, j, d, m, t, f	Individuals, products, demographic groups, municipalities, markets (years), firms
P	In-person channel
O	Online channel
M_{dmt}	Number of individuals in municipality m, demographic group d, and market t
M_{dt}	Number of individuals in demographic group d and market t
M_t	Number of individuals in market t
$\mathcal{M}$	Set of all municipalities
J_t	Number of products available in market t
$\mathcal{J}_t$	Set of products available in market t
$\mathcal{F}$	Set of all firms
$\mathcal{G}_{ft}$	Set of products offered by firm f in market t
D	Number of demographic groups
T	Number of markets (years)
U_{ijdm}	Indirect utility of consumer i of group d in municipality m for product j
X_j	Vector of observed car characteristics of product j
ξ_{jd}	Unobserved demand residual of product j for group d
δ_{jd}	Mean indirect utility of product j for group d
dem_{dm}	Vector of observable average demographics for group d in municipality m
ν_i	Vector of random coefficients for consumer i
ϵ_{ijdm}	Idiosyncratic taste shock of consumer i of group d in municipality m for product j
α_d	Price sensitivity of group d
β_d	Preference parameters for car characteristics of group d
γ_d	Search cost sensitivity of group d
Π_d	Parameters for demographic interactions of group d
Σ_d	Parameters for random coefficients of group d
θ_d	Set of all demand-side parameters $(\beta_d, \alpha_d, \Pi_d, \Sigma_d, \gamma_d, \tau_d)$
λ	Cost function parameters (λ_1, λ_2)
sc_{ifdm}	Search cost for consumer i of group d in municipality m visiting the closest dealer of firm f
$\overline{\text{sc}}^f_{idm}$	Expected search cost, $\mathbb{E}[\text{sc}_{ifdm}]$
dist_{fm}	Average driving distance from municipality m to the closest dealer of firm f
ψ_d	Share of consumers of group d willing to purchase cars online
τ_d	Factor of search cost reduction for online sales, $\tau_d \in [0, 1]$ (Tesla)
ρ	Factor of search cost reduction for online sales, $\rho \in [0, 1]$ (counterfactuals)
p_{jdt}	Transaction price of product j for group d in market t (discriminatory price)
$\overline{p}_{jt}$	List price of product j in market t
c_{jt}	Marginal cost of product j in market t
W_j	Vector of cost shifters for product j
ω_j	Unobserved cost shock of product j
s_{jdmt}	Market share of product j for group d in municipality m and market t
s_{jdt}	Market share of product j for group d in market t
p^P_{jdt}	In-person price of product j for group d in market t
p^O_{jt}	Online (uniform) price of product j in market t
s^P_{jdmt}	In-person market share of product j for group d in municipality m and market t
s^O_{jdmt}	Online market share of product j for group d in municipality m and market t
s^P_{jdt}	In-person market share of product j for group d in market t
s^O_{jdt}	Online market share of product j for group d in market t
ϕ_{dt}	Share of consumers in market t that belong to demographic group d (M_{dt}/M_t)
w_{dmt}	Share of consumers in demographic group d and market t that live in municipality m (M_{dmt}/M_{dt})

Table A.9: Specification details

Variable	Description
Car characteristics	
Price	Price, net of French feebate program, in 10,000 2018 euros (Demand only)
Distance	Driving distance, in 10km (Demand only)
Horsepower	Horsepower, in 100kW
Weight	Curb weight, in 1,000kg
Fuel cost	Cost for driving 100km, in 2018 euros (Demand only)
Fuel consumption	Fuel consumption, in L / 100km (Supply only)
Diesel	=1 if Diesel
Electric	=1 if Electric
Plug-in hybrid	=1 if Plug-in hybrid
Hybrid	=1 if Hybrid
Station wagon	=1 if Station Wagon
Convertible	=1 if Convertible
Trend	Time trend (Supply only)
Demographics (Demand only)	
Median income	Median income, in 10,000 2018 euros
Average age	Average age, in 10 years
Share of women	Share of women, in percentage
Household size	Population / Number of households
Urban	=1 if Urban
	Notes: All demographics are demeaned and are at demographic group-by-municipality level.
Cost shifters (Supply only)	
Input price index	Composite price index based on steel price (77%), polypropylene price (12%), and aluminum price (11%), interacted with vehicle weight (Supply only)
Real exchange rate	Penn World Table 10.0, `pl_con`, see Grieco et al. (2023) (Supply only)
	Notes: Both cost shifters are lagged one period to reflect planning horizons.
Instruments	
Demand-side	(1) Sum of characteristics of competitors using horsepower, weight, fuel cost
	(2) Number of competitors' products
	(3) Number of competitors' products with same engine type
	(4) Number of competitors' products with same body trim
	Note: Demand-side instruments are the same for all demographic groups.
Supply-side	(1) Sum of characteristics of competitors using horsepower, weight, fuel consumption, input price index, real exchange rate
	(2) Number of competitors' products
	(3) Number of competitors' products with same engine type
	(4) Number of competitors' products with same body trim

$= (g_{31}(\Pi_1, \Sigma_1, \gamma_1, \tau_1), \ldots, g_{3D}(\Pi_D, \Sigma_D, \gamma_D, \tau_D))$ as defined in (18).

The purchase probabilities predicted by the model are obtained by evaluating (7) at any given $(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$: $s_{jdm}(\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d), \Pi_d, \Sigma_d, \gamma_d, \tau_d) = s_{jdm}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$, where we obtain $\delta_d(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ by inverting the system of J national-level market shares in (8). We perform this demand inverse using the SQUAREM algorithm, see Varadhan and Roland (2008), and a tight convergence threshold of `1e-14`.

The integral in equation (7) is of dimension one, as we include only a scalar random coefficient on the intercept (common to all $j \neq 0$) while the demographic characteristics

dem_{dm} and the distance dist_{fm} are observed. We approximate the integral in (7) using a Gauss-Hermite quadrature with 15 nodes (see Liu and Pierce, 1994).

The micro moments $g_{3d}(\Pi_d, \Sigma_d, \gamma_d, \tau_d)$ are computed using a random sample of 3,000 municipalities (per demographic group), representing about 10% of all municipalities in France. To avoid sampling the same municipality twice and preserve regional representativeness, we draw the sample of municipalities using systematic sampling and replace the population weights w_{dm} in (18) with appropriate weights $\widetilde{w}_{dm}$ that account for our sampling procedure.

The GMM estimator for this first step is

$$(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) = \underset{\Pi,\Sigma,\gamma,\tau}{\arg\min} \quad g_3(\Pi, \Sigma, \gamma, \tau)' \, \mathbf{W} \, g_3(\Pi, \Sigma, \gamma, \tau), \tag{37}$$

where $\mathbf{W}$ is an appropriate weighting matrix. We first obtain consistent (but potentially inefficient) estimates of $(\Pi, \Sigma, \gamma, \tau)$, say $(\widetilde{\Pi}, \widetilde{\Sigma}, \widetilde{\gamma}, \widetilde{\tau})$, using the identity matrix as the weighting matrix, then we compute the optimal weighting matrix $\mathbf{W}$ based on the preliminary estimates $(\widetilde{\Pi}, \widetilde{\Sigma}, \widetilde{\gamma}, \widetilde{\tau})$ following the best practices described in Conlon and Gortmaker (2025), and finally recalculate (37) to obtain $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$.

Our final specification includes distance, income, household size, an urban indicator, and interactions between income and horsepower, urban and horsepower, and urban and weight. We include the associated micro moments for all car models as well as Tesla-specific micro moments for distance, income, household size, and the urban indicator. We define the micro moments at the demographic group-by-year level, so our first-step GMM estimator includes 858 moments (11 "types" of moments $\times$ 6 groups $\times$ 13 years).

Our first-step GMM yields the parameter estimates $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$ and the average group-specific indirect utilities $(\delta_{jd}(\widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d))_{j=1,\dots,J,\, d=1,\dots,D}$. We take these parameter estimates as given in the second-step GMM estimator we describe next.

Estimation of the linear parameters. In the second-step GMM estimator of the linear parameters (α, β, λ), we concentrate out (β, λ) to simplify implementation. We perform the following steps. For given $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$ and a value of the price coefficients α, we calculate the marginal costs and the transaction prices from the FOCs of the producers:

$$c_j(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) = \overline{p}_j + \min_d \left\{ \left[\widetilde{\mathcal{D}}_d(\alpha_d, \widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d)^{-1} \cdot s_d(\widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d) \right]_j \right\}$$

and

$$p_{jd}(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) = c_j(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) - \left[\widetilde{\mathcal{D}}_d(\alpha_d, \widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d)^{-1} \cdot s_d(\widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d)\right]_j,$$

which do not depend on β and λ. Given these and $(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$, we calculate $\widehat{\beta}(\alpha) = (\widehat{\beta}_d(\alpha))_{d=1,...,D}$ and $\widehat{\lambda}(\alpha)$ from the following OLS regressions:

$$\delta_{jd}(\widehat{\Pi}_d, \widehat{\Sigma}_d, \widehat{\gamma}_d, \widehat{\tau}_d) - \alpha_d\, p_{jd}(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) = X_j'\beta_d + \xi_{jd}$$

and

$$\ln\left(c_j(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})\right) = X_j'\lambda_1 + W_j'\lambda_2 + \omega_j.$$

Given the residuals of these OLS regressions, we then construct the following demand-side moment conditions:

$$g_{1d}(\widehat{\beta}_d(\alpha), \alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}) = \frac{1}{J}\sum_{j=1}^{J} Z_{jd}'\,\xi_{jd}(\widehat{\beta}_d(\alpha), \alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$$

for $d = 1, ..., D$, and supply-side moment conditions:

$$g_2(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}, \widehat{\lambda}(\alpha)) = \frac{1}{J}\sum_{j=1}^{J} Z_{jS}'\,\omega_j(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}, \widehat{\lambda}(\alpha)).$$

Denote the stacked vector of demand- and supply-side moments as $g(\alpha, \widehat{\beta}(\alpha), \widehat{\lambda}(\alpha)) \equiv (g_{11}(\widehat{\beta}_1(\alpha), \alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})',\ ...,\ g_{1D}(\widehat{\beta}_D(\alpha), \alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})',\ g_2(\alpha, \widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau}, \widehat{\lambda}(\alpha))')'$. Finally, our second-step GMM estimator is

$$\widehat{\alpha} = \arg\min_{\alpha} \quad g(\alpha, \widehat{\beta}(\alpha), \widehat{\lambda}(\alpha))'\,\mathbf{W}\,g(\alpha, \widehat{\beta}(\alpha), \widehat{\lambda}(\alpha)),$$

where $\mathbf{W} = \text{diag}(\mathbf{W}_1, ..., \mathbf{W}_D, \mathbf{W}_S)$ is a block-diagonal weighting matrix. In practice, since our instruments do not vary across demographic groups ($Z_{jd} = Z_j$ for all d), we define $\mathbf{W}_1 = ... = \mathbf{W}_D = (Z'\,Z)^{-1}$ for the demand-side moments and $\mathbf{W}_S = (Z_S'\,Z_S)^{-1}$ for the supply-side moments, where $Z = (Z_j)_{j=1,...,J}$ and $Z_S = (Z_{jS})_{j=1,...,J}$.

As explained in Section 3, we account for the estimation error in $(\widehat{\Pi}, \widehat{\Sigma}, \widehat{\gamma}, \widehat{\tau})$ arising from the first-step GMM estimator by implementing the variance-covariance correction in Newey and McFadden (1994).

D Price discrimination in the online channel

We consider counterfactuals in which firms can price discriminate in both the online and in-person distribution channels. While this does not align with firms' stated intentions about online sales (see Introduction), we acknowledge that they could in principle price discriminate also online, as data on consumers are readily available online. For example, consumer demographics could be inferred from browsing histories by third-party data brokers and resold to car manufacturers. The results are presented in Table A.10.

Table A.10: Online and offline price discrimination

	No online access	Limited online access		Universal online access	
		-37%	-100%	-37%	-100%
Panel A: Transaction prices, in-person channel					
Group 1: Young/Poor	21,418	21,395	21,396	21,405	21,405
Group 2: Young/Rich	21,665	21,629	21,629	21,645	21,645
Group 3: Middle/Poor	22,097	22,037	22,040	22,073	22,073
Group 4: Middle/Rich	22,750	22,651	22,653	22,708	22,708
Group 5: Old/Poor	22,099	22,014	22,018	22,082	22,082
Group 6: Old/Rich	23,733	23,565	23,569	23,677	23,677
Panel B: Transaction prices, online channel					
Group 1: Young/Poor	–	21,395	21,396	21,405	21,405
Group 2: Young/Rich	–	21,629	21,629	21,645	21,645
Group 3: Middle/Poor	–	22,037	22,040	22,073	22,073
Group 4: Middle/Rich	–	22,651	22,653	22,708	22,708
Group 5: Old/Poor	–	22,014	22,018	22,082	22,082
Group 6: Old/Rich	–	23,565	23,569	23,677	23,677

Notes: All counterfactual experiments are computed using the 2019 data only. The baseline is the observed market outcome (column 1). This table presents the results from a counterfactual experiment in which all manufacturers can price discriminate both in person and online. All monetary values are in 2018 euros. Transaction prices use uniform weights $w_j = \sum_d \phi_d s_{jd} / \sum_j \sum_d \phi_d s_{jd}$, constructed from total sales in the baseline scenario and fixed across demographic groups and counterfactual experiments.

The central result from Table A.10 is that, when price discrimination is allowed in both channels, firms optimally set the same prices online and in person across all counterfactual experiments: Panels A and B are identical in every column. Intuitively, once the competitive constraint imposed by uniform online pricing is removed, there is no incentive for firms to post different prices across distribution channels.

Compared to the baseline, the introduction of the online channel still leads to price reductions across all demographic groups, reflecting the competitive pressure associated with lower search costs. The reductions range from around €23 (young/poor) to €168 (old/rich) under the limited access case with a 37% search cost reduction, equivalent to the search cost advantage estimated from Tesla's sales.

Within the counterfactual experiments, further reducing search costs from 37% to 100% has virtually no effect on prices: comparing columns (2) and (3), or columns (4) and (5), prices change by at most €4 across all groups. This contrasts with our main results in Table 8, where search cost reductions generate more meaningful price changes. The difference arises because the key competitive mechanism in our main counterfactuals—the pricing discipline imposed by a uniform online price—is absent here.

Expanding consumer access leads to slight price increases across all groups, ranging from €9 (young/poor) to €112 (old/rich) at the 37% search cost reduction, and from €9 to €108 when search costs are eliminated entirely, with larger increases for less price-elastic groups. This price response is due to the market expansion effect that follows from eliminating search costs for a large number of consumers. When all consumers gain access to the online channel, consumers that were formerly unwilling to make an online purchase can now search across both channels at lower cost, increasing market demand. Since firms already set identical prices across channels, as shown in Table A.10, this broader market participation does not translate into additional competitive pressure through channel switching. Instead, firms respond by slightly raising prices for all groups to extract part of the surplus generated from the lower aggregate search costs.